VERSION 1 SANK INTO THE SHARK RIVER

STARTING FRESH WITH VERSION 2:

A) - 2 PROCESSORS: (LIKE BEFORE)

1) TO CONTROL MESHTASTIC
   - A HELTEC V4 RADIO + CONTROLLER

2) AN ADAFRUIT FEATHER V2 (SAME AS BEFORE)
   TO CONTROL ACOUSTIC DISTANCE SENSOR

B) SKIP BME 280 TPH SENSOR FOR NOW

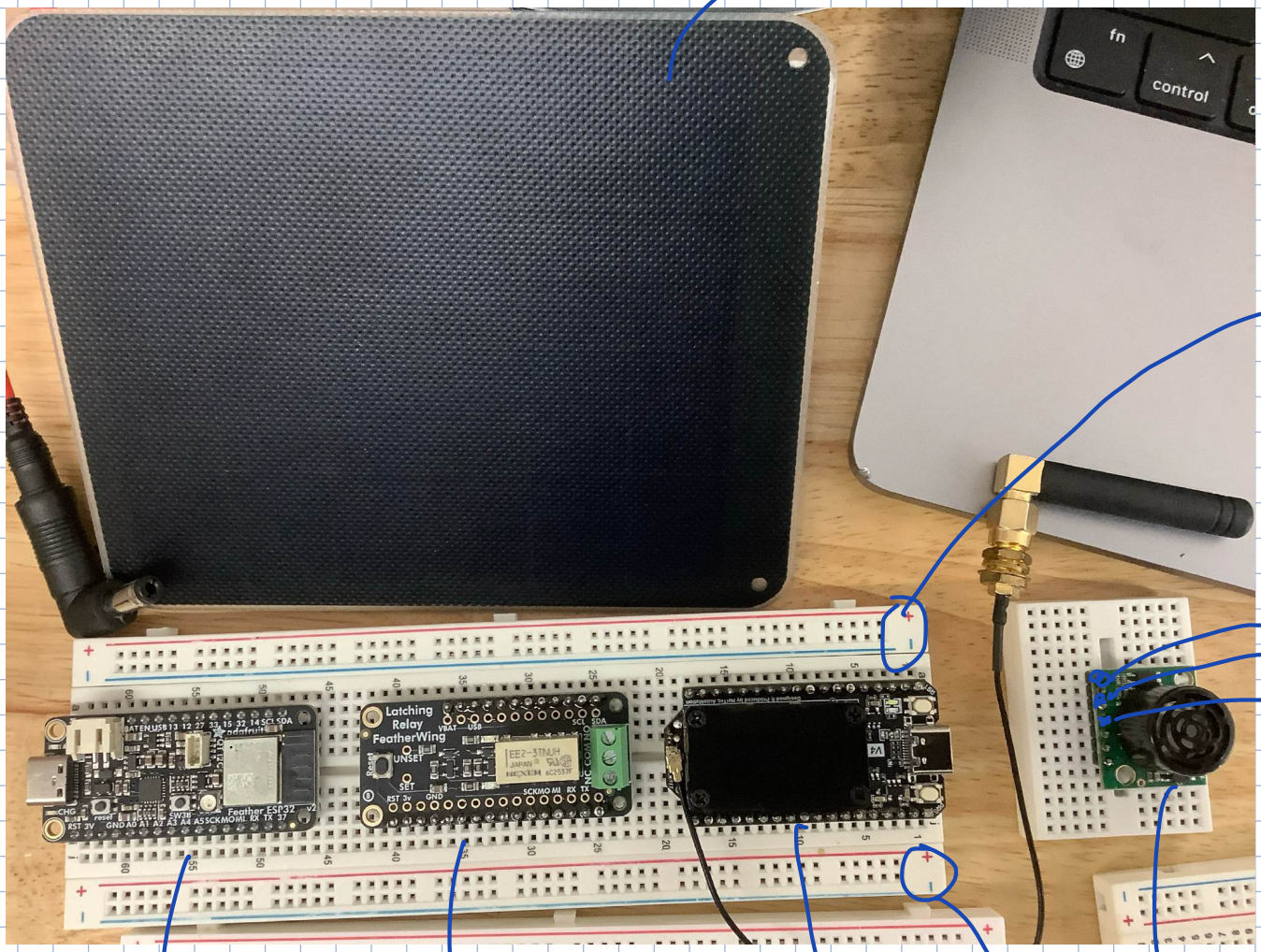
C) NEW ACOUSTIC DISTANCE SENSOR
   ~ MAXBOTIX ULTRASONIC FROM ADAFRUIT
   - RECOMMENDED ON YOUTUBE TO BE MORE ACCURATE + ROBUST THAN OLD SENSOR

D) EVERYTHING CAN RUN OFF 3.3 V
   (I THINK)

E) CAN'T FIND MY SOLAR BATTERY CHARGER CHIP ORDERED FROM ADAFRUIT IN APRIL
   * - I ORDERED ANOTHER ONE, PLUS A CONNECTOR TO SOLAR PANEL (2W) AND A NEW BME 280 TPH SENSOR

F) DO I NEED LATCHING RELAY? CAN I TURN EVERYTHING ELSE OFF WITH FEATHER SLEEP MODE?

POWER EVERYTHING WITH FEATHER USB C PORT FOR NOW UNTIL CHARGER BOARD ARRIVES

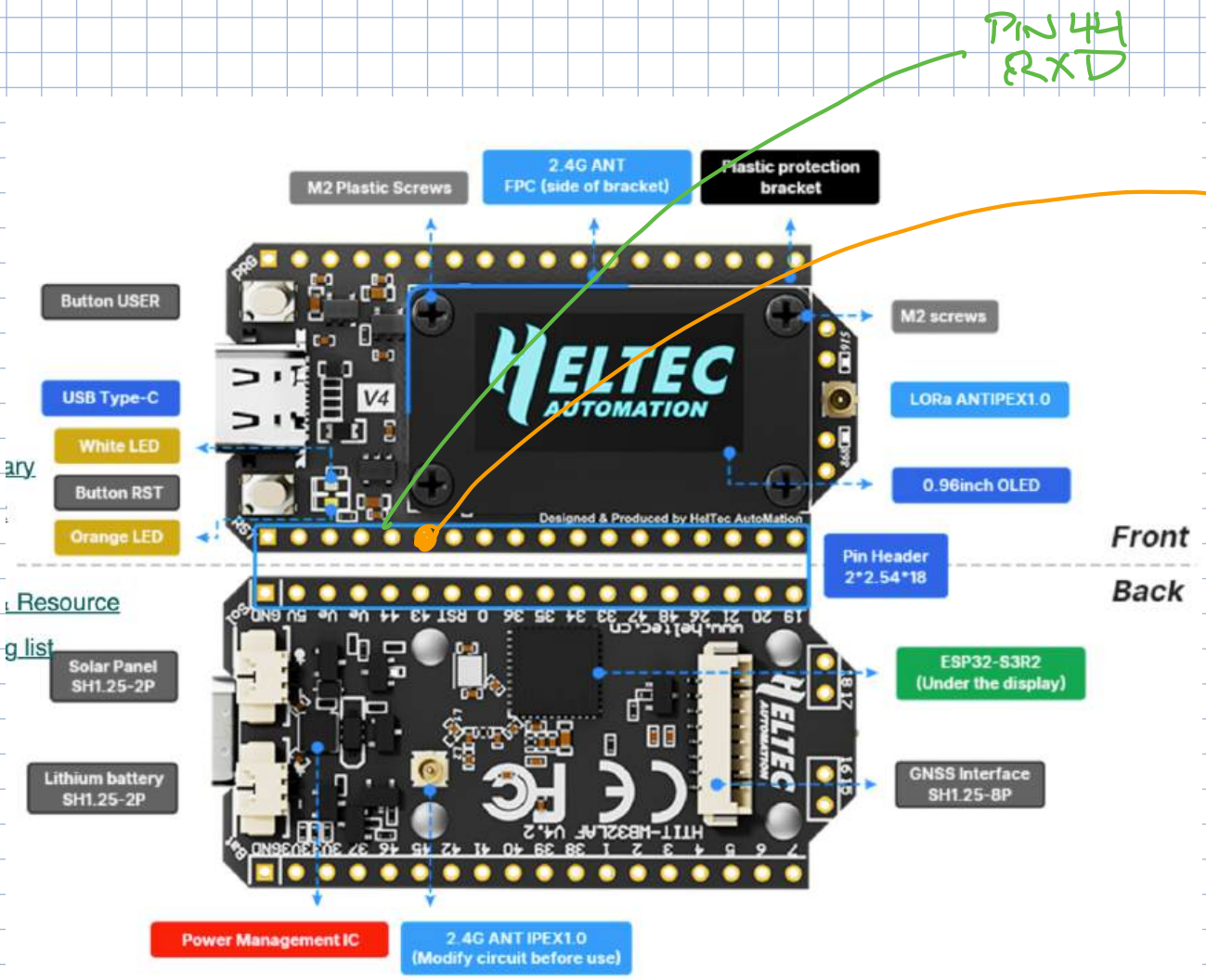


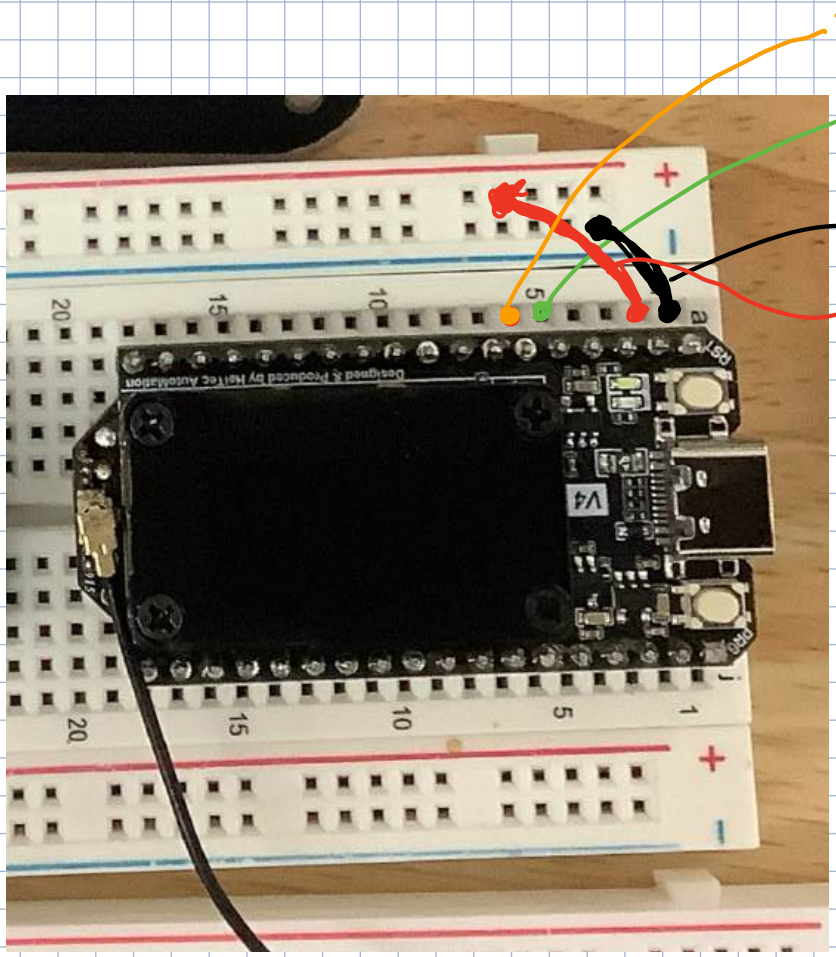
(DON'T USE FOR NOW - I THINK I CAN SWITCH EVERYTHING OFF WITH FEATHER

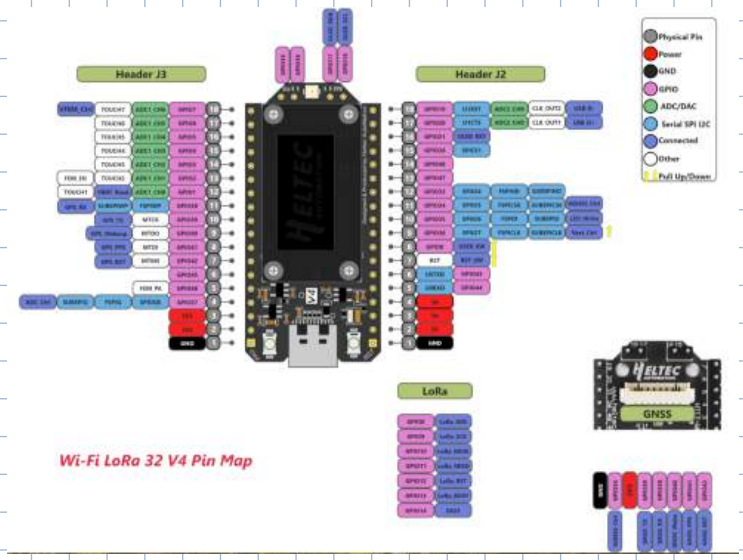
1ST INSTALL MESHTASTIC SOFTWARE WITH SERIAL MODULE

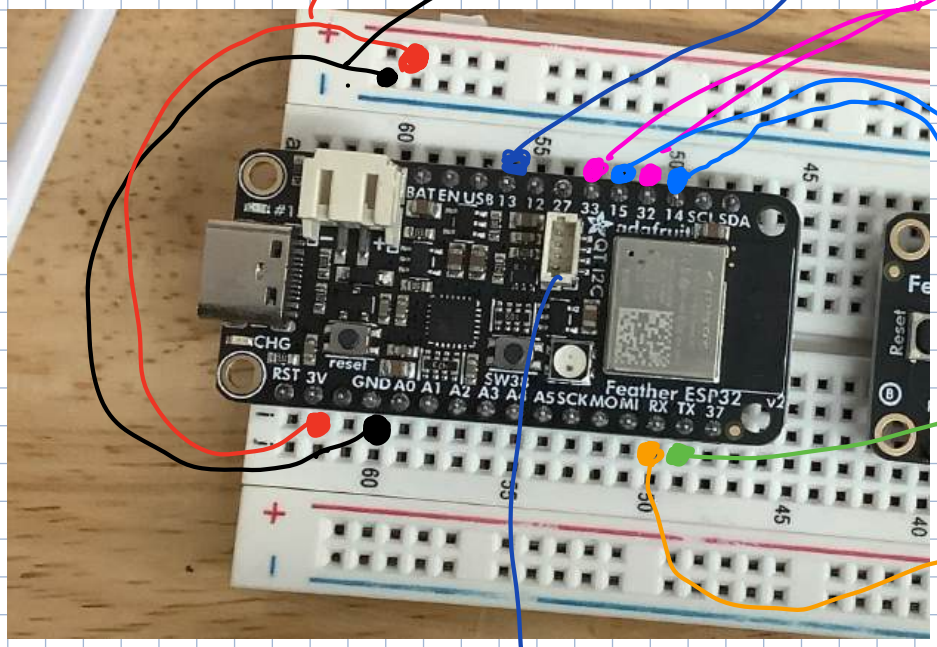
- OK

RXD PIN 44
TXD PIN 43


PIN 44
RXD
PIN 43
TXD
M2 Plastic Screws
2.4G ANT
FPC (side of bracket)
Plastic protection
bracket
Button USER
USB Type-C
White LED
Button RST
Orange LED
M2 screws
LORa ANTIPEX1.0
0.96inch OLED
Front
Back
Pin Header
2*2.54*18
Solar Panel
SH1.25-2P
ESP32-S3R2
(Under the display)
Lithium battery
SH1.25-2P
GNSS Interface
SH1.25-8P
Power Management IC
2.4G ANT IPEX1.0
(Modify circuit before use)
HELTEC
AUTOMATION
V4
Designed & Produced by HelTec AutoMation


PIN 43
TXD
PIN 44
RXD
GND
3.3V
POWER
HELTEC V4
CHIP


Header J3
Header J2
LoRa
GNSS
Wi-Fi LoRa 32 V4 Pin Map



- GND
- 3.3V POWER
- RX HELTEC TO TX FEATHER
- TX HELTEC TO RX FEATHER

(SERIAL 1 ON FEATHER ARDUINO)

LV-MaxSonar®-EZ™ Series

High Performance Sonar Range Finder

MB1000, MB1010, MB1020, MB1030, MB1040[2]



With 2.5V - 5.5V power the LV-MaxSonar-EZ provides very short to long-range detection and ranging in a very small package. The LV-MaxSonar-EZ detects objects from 0-inches to 254-inches (6.45-meters) and provides sonar range information from 6-inches out to 254-inches with 1-inch resolution. Objects from 0-inches to 6-inches typically range as 6-inches[1]. The interface output formats included are pulse width output, analog voltage output, and RS232 serial output. Factory calibration and testing is completed with a flat object. ***[1]See Close Range Operation***

Features

- Continuously variable gain for control and side lobe suppression
- Object detection to zero range objects
- 2.5V to 5.5V supply with 2mA typical current draw
- Readings can occur up to every 50mS, (20-Hz rate)
- Free run operation can continually measure and output range information
- Triggered operation provides the range reading as desired
- Interfaces are active simultaneously
- Serial, 0 to Vcc, 9600 Baud, 81N
- Analog, (Vcc/512) / inch
- Pulse width, (147uS/inch)
- Learns ringdown pattern when commanded to start ranging
- Designed for protected indoor environments
- Sensor operates at 42KHz
- High output square wave sensor drive (double Vcc)
- Actual operating temperature range from –40°C to +65°C, Recommended operating temperature range from 0°C to +60°C [1]

Benefits

- Very low cost ultrasonic rangefinder
- Reliable and stable range data
- Quality beam characteristics
- Mounting holes provided on the circuit board
- Very low power ranger, excellent for multiple sensor or battery-based systems
- Fast measurement cycles
- Sensor reports the range reading directly and frees up user processor
- Choose one of three sensor outputs
- Triggered externally or internally

Applications and Uses

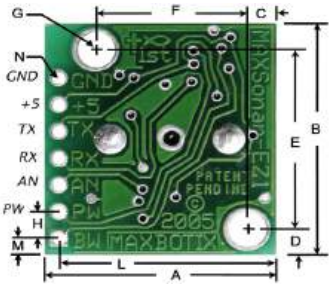
- UAV blimps, micro planes and some helicopters
- Bin level measurement
- Proximity zone detection
- People detection
- Robot ranging sensor
- Autonomous navigation
- Multi-sensor arrays
- Distance measuring
- Long range object detection
- Wide beam sensitivity

Notes:

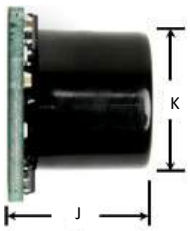
[1]Please reference page 4 for minimum operating voltage verses temperature information.

[2] Please reference page 12 for part number key.

LV-MaxSonar-EZ Mechanical Dimensions



A	0.785"	19.9 mm	H	0.100"	2.54 mm
B	0.870"	22.1 mm	J	0.610"	15.5 mm
C	0.100"	2.54 mm	K	0.645"	16.4 mm
D	0.100"	2.54 mm	L	0.735"	18.7 mm
E	0.670"	17.0 mm	M	0.065"	1.7 mm
F	0.510"	12.6 mm	N	0.038" dia.	1.0 mm dia.
G	0.124" dia.	3.1 mm dia.	weight, 4.3 grams		



Close Range Operation

Applications requiring 100% reading-to-reading reliability should not use MaxSonar sensors at a distance closer than 6 inches. Although most users find MaxSonar sensors to work reliably from 0 to 6 inches for detecting objects in many applications, MaxBotix® Inc. does not guarantee operational reliability for objects closer than the minimum reported distance. Because of ultrasonic physics, these sensors are unable to achieve 100% reliability at close distances.

Warning: Personal Safety Applications

We do not recommend or endorse this product be used as a component in any personal safety applications. This product is not designed, intended or authorized for such use. These sensors and controls do not include the self-checking redundant circuitry needed for such use. Such unauthorized use may create a failure of the MaxBotix® Inc. product which may result in personal injury or death. MaxBotix® Inc. will not be held liable for unauthorized use of this component.

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About Ultrasonic Sensors

Our ultrasonic sensors are in air, non-contact object detection and ranging sensors that detect objects within an area. These sensors are not affected by the color or other visual characteristics of the detected object. Ultrasonic sensors use high frequency sound to detect and localize objects in a variety of environments. Ultrasonic sensors measure the time of flight for sound that has been transmitted to and reflected back from nearby objects. Based upon the time of flight, the sensor then outputs a range reading.

Pin Out Description

Pin 1-BW-*Leave open or hold low for serial output on the TX output. When BW pin is held high the TX output sends a pulse (instead of serial data), suitable for low noise chaining.

Pin 2-PW- This pin outputs a pulse width representation of range. The distance can be calculated using the scale factor of 147uS per inch.

Pin 3-AN- Outputs analog voltage with a scaling factor of (Vcc/512) per inch. A supply of 5V yields ~9.8mV/in. and 3.3V yields ~6.4mV/in. The output is buffered and corresponds to the most recent range data.

Pin 4-RX– This pin is internally pulled high. The LV-MaxSonar-EZ will continually measure range and output if RX data is left unconnected or held high. If held low the sensor will stop ranging. Bring high for 20uS or more to command a range reading.

Pin 5-TX- When the *BW is open or held low, the TX output delivers asynchronous serial with an RS232 format, except voltages are 0-Vcc. The output is an ASCII capital "R", followed by three ASCII character digits representing the range in inches up to a maximum of 255, followed by a carriage return (ASCII 13). The baud rate is 9600, 8 bits, no parity, with one stop bit. Although the voltage of 0-Vcc is outside the RS232 standard, most RS232 devices have sufficient margin to read 0-Vcc serial data. If standard voltage level RS232 is desired, invert, and connect an RS232 converter such as a MAX232. When BW pin is held high the TX output sends a single pulse, suitable for low noise chaining. (no serial data)

Pin 6-+5V- Vcc – Operates on 2.5V - 5.5V. Recommended current capability of 3mA for 5V, and 2mA for 3V. Please reference page 4 for minimum operating voltage verses temperature information.

Pin 7-GND- Return for the DC power supply. GND (& Vcc) must be ripple and noise free for best operation.

Range "0" Location

The LV-MaxSonar-EZ reports the range to distant targets starting from the front of the sensor as shown in the diagram below.



Range Zero

The range is measured from the front of the transducer.

In general, the LV-MaxSonar-EZ will report the range to the leading edge of the closest detectable object. Target detection has been characterized in the sensor beam patterns.

Sensor Minimum Distance

The sensor minimum reported distance is 6-inches (15.2 cm). However, the LV-MaxSonar-EZ will range and report targets to the front sensor face. Large targets closer than 6-inches will typically range as 6-inches.

Sensor Operation from 6-inches to 20-inches

Because of acoustic phase effects in the near field, objects between 6-inches and 20-inches may experience acoustic phase cancellation of the returning waveform resulting in inaccuracies of up to 2-inches. These effects become less prevalent as the target distance increases, and has not been observed past 20-inches.

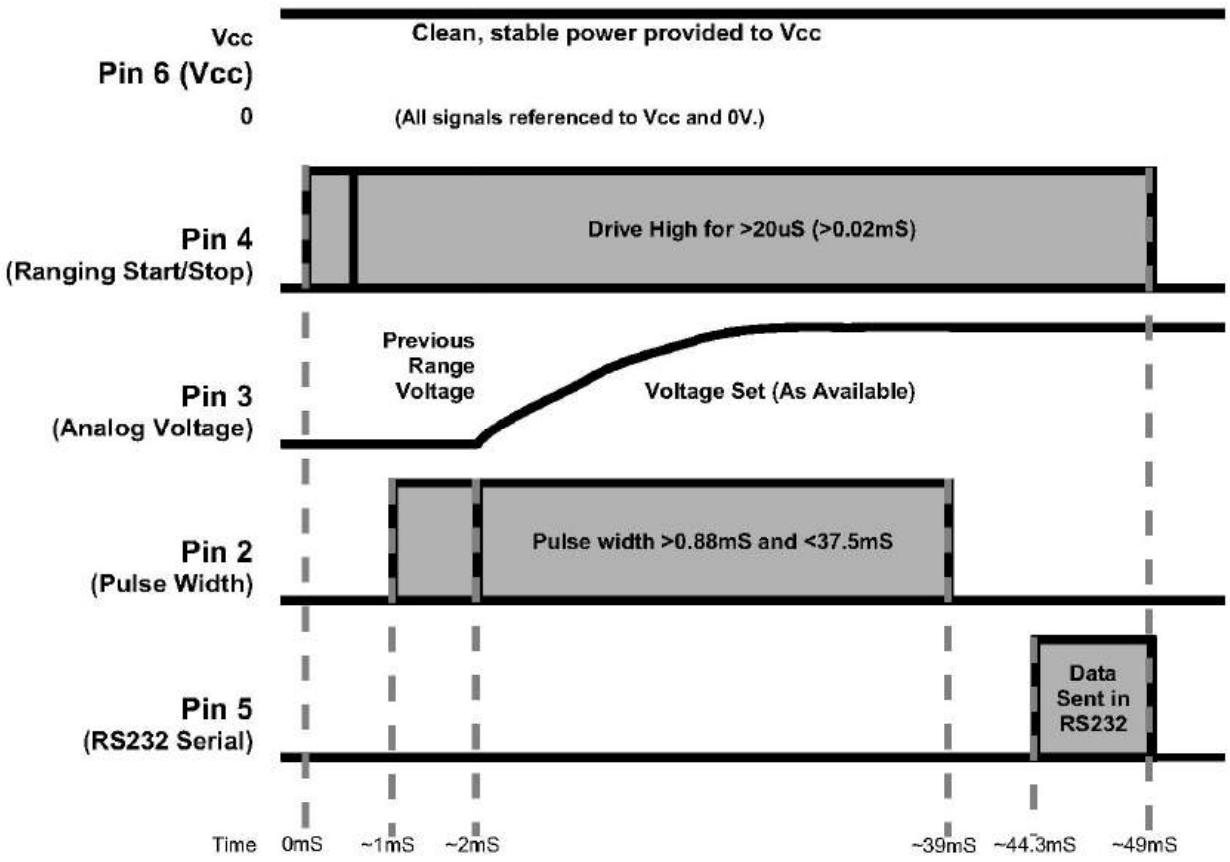
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General Power-Up Instruction

Each time the LV-MaxSonar-EZ is powered up, it will calibrate during its first read cycle. The sensor uses this stored information to range a close object. It is important that objects not be close to the sensor during this calibration cycle. The best sensitivity is obtained when the detection area is clear for fourteen inches, but good results are common when clear for at least seven inches. If an object is too close during the calibration cycle, the sensor may ignore objects at that distance.

The LV-MaxSonar-EZ does not use the calibration data to temperature compensate for range, but instead to compensate for the sensor ringdown pattern. If the temperature, humidity, or applied voltage changes during operation, the sensor may require recalibration to reacquire the ringdown pattern. Unless recalibrated, if the temperature increases, the sensor is more likely to have false close readings. If the temperature decreases, the sensor is more likely to have reduced up close sensitivity. To recalibrate the LV-MaxSonar-EZ, cycle power, then command a read cycle.

Timing Diagram



Timing Description

250mS after power-up, the LV-MaxSonar-EZ is ready to accept the RX command. If the RX pin is left open or held high, the sensor will first run a calibration cycle (49mS), and then it will take a range reading (49mS). After the power up delay, the first reading will take an additional ~100mS. Subsequent readings will take 49mS. The LV-MaxSonar-EZ checks the RX pin at the end of every cycle. Range data can be acquired once every 49mS.

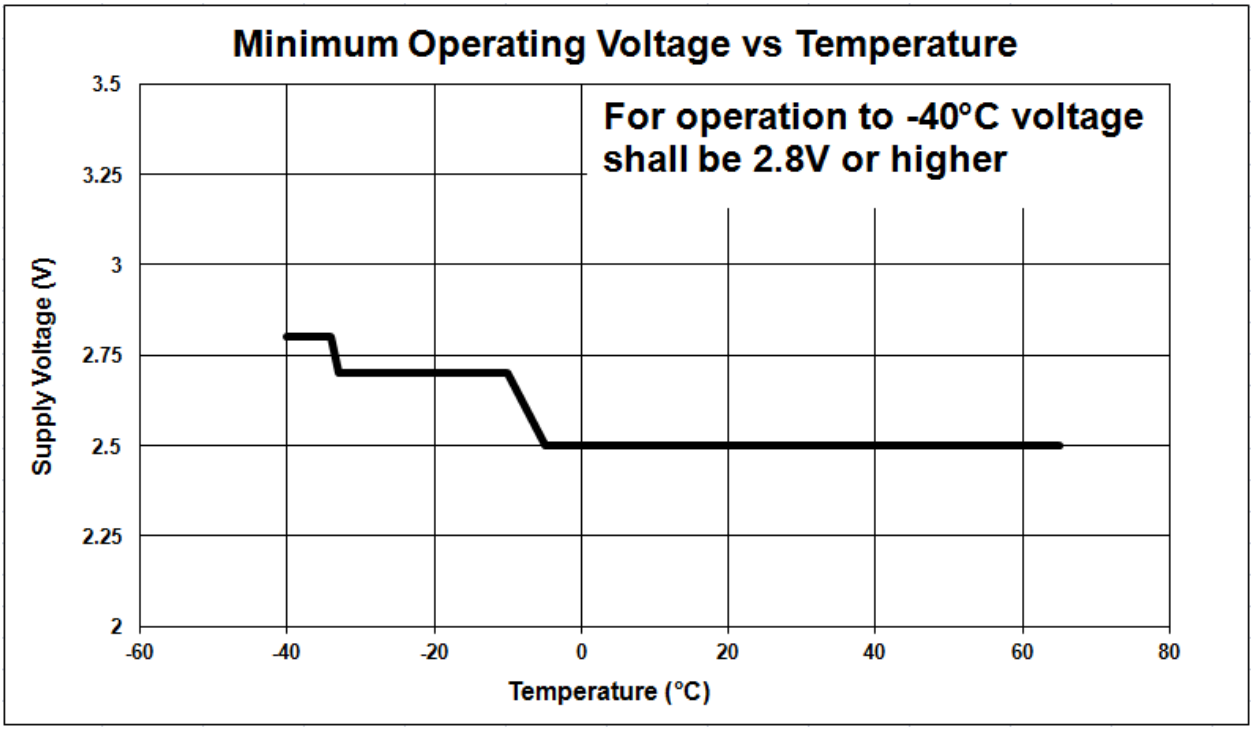
Each 49mS period starts by the RX being high or open, after which the LV-MaxSonar-EZ sends the transmit burst, after which the pulse width pin (PW) is set high. When a target is detected the PW pin is pulled low. The PW pin is high for up to 37.5mS if no target is detected. The remainder of the 49mS time (less 4.7mS) is spent adjusting the analog voltage to the correct level. When a long distance is measured immediately after a short distance reading, the analog voltage may not reach the exact level within one read cycle. During the last 4.7mS, the serial data is sent.

The LV-MaxSonar-EZ timing is factory calibrated to one percent at five volts, and in use is better than two percent. In addition, operation at 3.3V typically causes the objects range, to be reported, one to two percent further than actual.

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Voltage vs Temperature

The graph below shows minimum operating voltage of the sensor verses temperature.

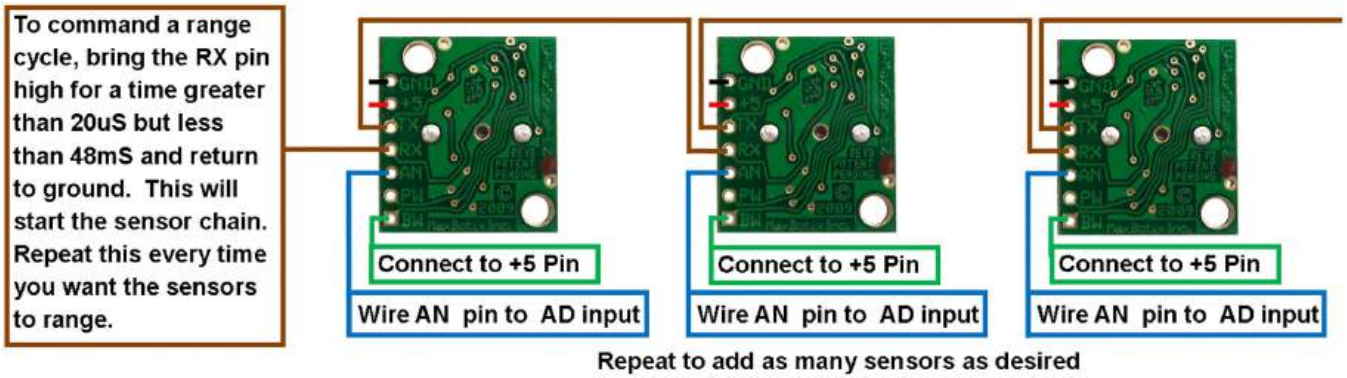


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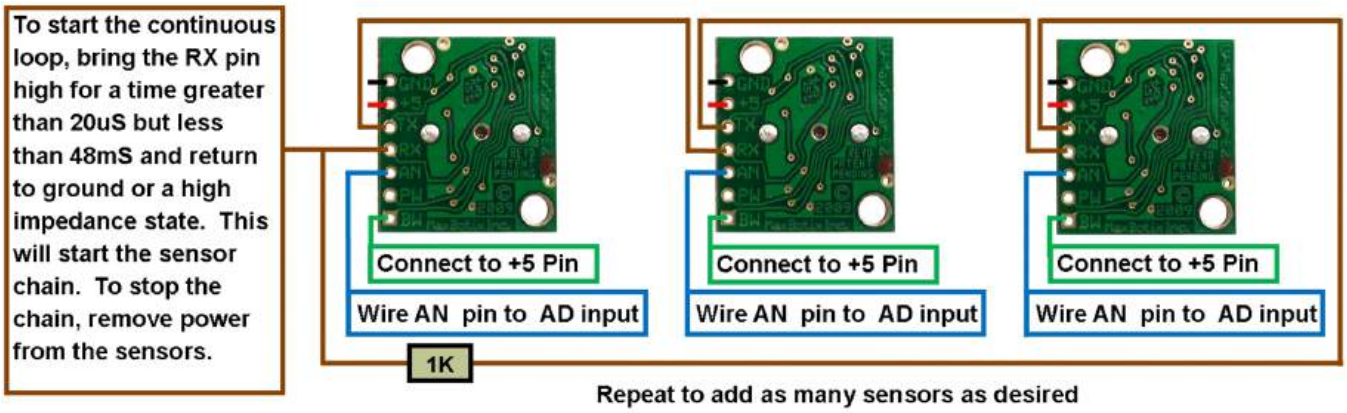
Using Multiple Sensors in a single system

When using multiple ultrasonic sensors in a single system, there can be interference (cross-talk) from the other sensors. MaxBotix Inc., has engineered and supplied a solution to this problem for the LV-MaxSonar-EZ sensors. The solution is referred to as chaining. We have 3 methods of chaining that work well to avoid the issue of cross-talk.

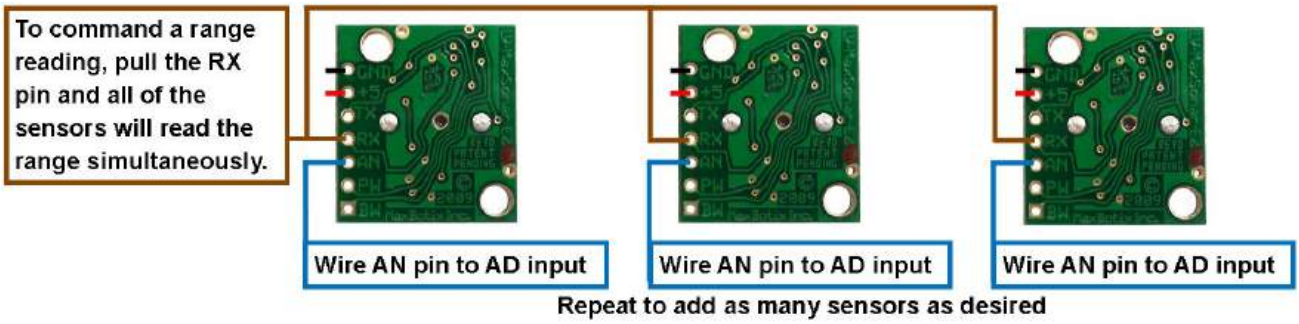
The first method is AN Output Commanded Loop. The first sensor will range, then trigger the next sensor to range and so on for all the sensor in the array. Once the last sensor has ranged, the array stops until the first sensor is triggered to range again. Below is a diagram on how to set this up.



The next method is AN Output Constantly Looping. The first sensor will range, then trigger the next sensor to range and so on for all the sensor in the array. Once the last sensor has ranged, it will trigger the first sensor in the array to range again and will continue this loop indefinitely. Below is a diagram on how to set this up.



The final method is AN Output Simultaneous Operation. This method does not work in all applications and is sensitive to how the other sensors in the array are positioned in comparison to each other. Testing is recommend to verify this method will work for your application. All the sensors RX pins are conned together and triggered at the same time causing all the sensor to take a range reading at the same time. Once the range reading is complete, the sensors stop ranging until triggered next time. Below is a diagram on how to set this up.

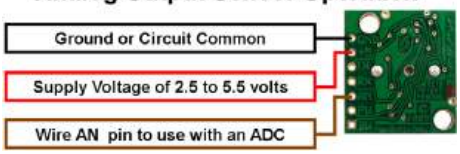


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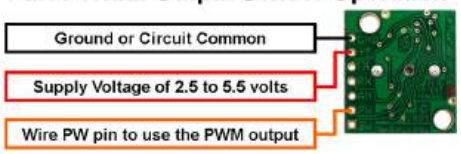
Independent Sensor Operation

The LV-MaxSonar-EZ sensors have the capability to operate independently when the user desires. When using the LV-MaxSonar-EZ sensors in single or independent sensor operation, it is easiest to allow the sensor to free-run. Free-run is the default mode of operation for all of the MaxBotix Inc., sensors. The LV-MaxSonar-EZ sensors have three separate outputs that update the range data simultaneously: Analog Voltage, Pulse Width, and RS232 Serial. Below are diagrams on how to connect the sensor for each of the three outputs when operating in a single or independent sensor operating environment.

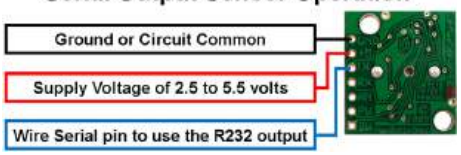
Analog Output Sensor Operation



Pulse Width Output Sensor Operation



Serial Output Sensor Operation



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Selecting an LV-MaxSonar-EZ

Different applications require different sensors. The LV-MaxSonar-EZ product line offers varied sensitivity to allow you to select the best sensor to meet your needs.

The diagram above shows how each product balances sensitivity and noise tolerance. This does not effect the maximum range, pin outputs, or other operations of the sensor. To view how each sensor will function to different sized targets reference the LV-MaxSonar-EZ Beam Patterns.

The LV-MaxSonar-EZ Sensors At a Glance

People Detection Wide Beam High Sensitivity		Best Balance		Large Targets Narrow Beam Noise Tolerance
MB1000	MB1010	MB1020	MB1030	MB1040

Background Information Regarding our Beam Patterns

Each LV-MaxSonar-EZ sensor has a calibrated beam pattern. Each sensor is matched to provide the approximate detection pattern shown in this datasheet. This allows end users to select the part number that matches their given sensing application. Each part number has a consistent field of detection so additional units of the same part number will have similar beam patterns. The beam plots are provided to help identify an estimated detection zone for an application based on the acoustic properties of a target versus the plotted beam patterns.

People Sensing: For users that desire to detect people, the detection area to the 1-inch diameter dowel, in general, represents the area that the sensor will reliably detect people.

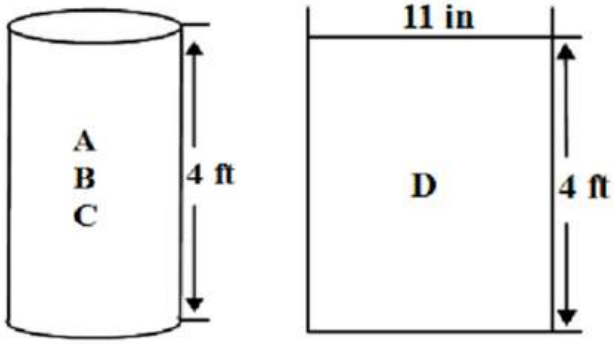
Each beam pattern is a 2D representation of the detection area of the sensor. The beam pattern is actually shaped like a 3D cone (having the same detection pattern both vertically and horizontally). Detection patterns for dowels are used to show the beam pattern of each sensor. Dowels are long cylindered targets of a given diameter. The dowels provide consistent target detection characteristics for a given size target which allows easy comparison of one MaxSonar sensor to another MaxSonar sensor.

For each part number, the four patterns (A, B, C, and D) represent the detection zone for a given target size. Each beam pattern shown is determined by the sensor's part number and target size.

The actual beam angle changes over the full range. Use the beam pattern for a specific target at any given distance to calculate the beam angle for that target at the specific distance. Generally, smaller targets are detected over a narrower beam angle and a shorter distance. Larger targets are detected over a wider beam angle and a longer range.

Beam Pattern Target Shapes

A 6.1-mm (0.25-inch) diameter dowel 4ft length
B 2.54-cm (1-inch) diameter dowel 4ft length
C 8.89-cm (3.5-inch) diameter dowel 4ft length
D 11-inch wide board 4ft in length moved left to right with the board parallel to the front sensor face. This shows the sensor's range capability.



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MB1000 LV-MaxSonar-EZ0

The LV-MaxSonar-EZ0 is the highest sensitivity and widest beam sensor of the LV-MaxSonar-EZ sensor series. The wide beam makes this sensor ideal for a variety of applications including people detection, autonomous navigation, and wide beam applications.

MB1000-000 MB1000-040

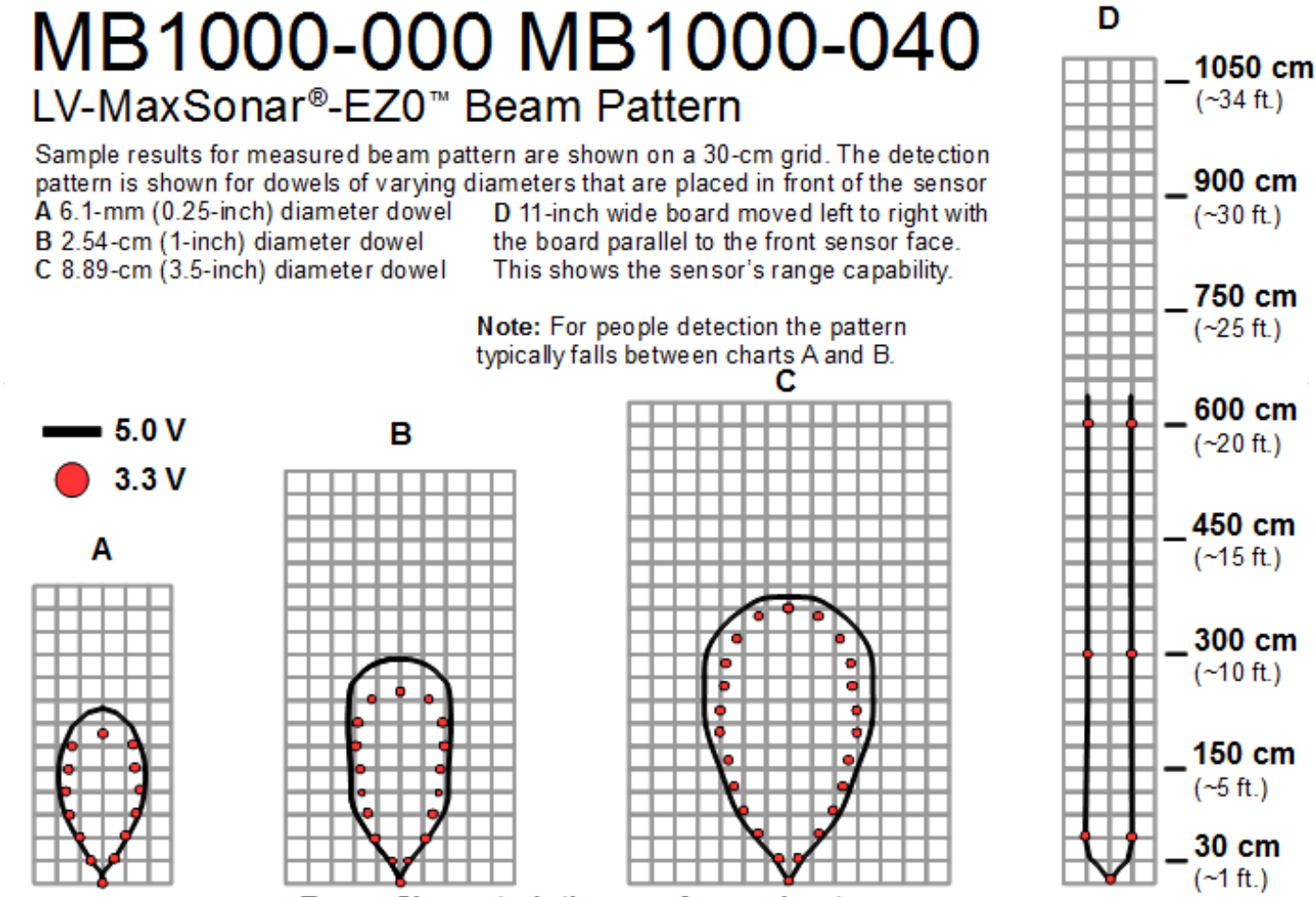
LV-MaxSonar®-EZ0™ Beam Pattern

Sample results for measured beam pattern are shown on a 30-cm grid. The detection pattern is shown for dowels of varying diameters that are placed in front of the sensor

A 6.1-mm (0.25-inch) diameter dowel
B 2.54-cm (1-inch) diameter dowel
C 8.89-cm (3.5-inch) diameter dowel

D 11-inch wide board moved left to right with the board parallel to the front sensor face. This shows the sensor's range capability.

Note: For people detection the pattern typically falls between charts A and B.



Beam Characteristics are Approximate

Beam Pattern drawn to a 1:95 scale for easy comparison to our other products.

MB1000 Features and Benefits

- Widest and most sensitive beam pattern in LV-MaxSonar-EZ line
- Low power consumption
- Easy to use interface
- Will pick up the most noise clutter when compared to other sensors in the LV-MaxSonar-EZ line
- Detects smaller objects
- Best sensor to detect soft object in LV-MaxSonar-EZ line
- Requires use of less sensors to cover a given area
- Can be powered by many different types of power sources
- Can detect people up to approximately 10 feet

MB1000 Applications and Uses

- Great for people detection
- Security
- Motion detection
- Used with battery power
- Autonomous navigation
- Educational and hobby robotics
- Collision avoidance

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MB1010 LV-MaxSonar-EZ1

The LV-MaxSonar-EZ1 is the original MaxSonar product. This is our most popular indoor ultrasonic sensor and is a great low-cost general-purpose sensor for a customer not sure of which LV-MaxSonar-EZ sensor to use.

MB1010-000 MB1010-040

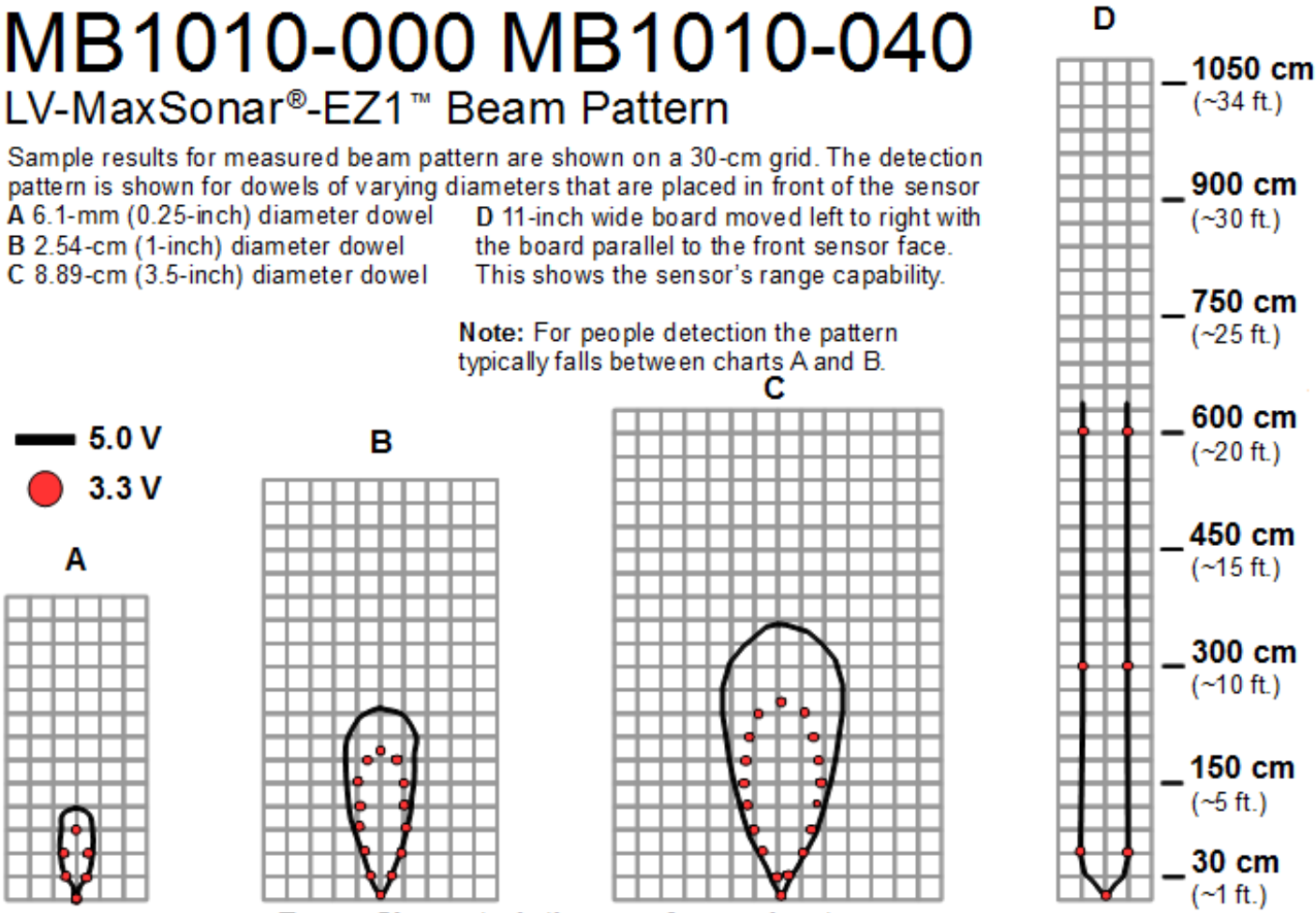
LV-MaxSonar®-EZ1™ Beam Pattern

Sample results for measured beam pattern are shown on a 30-cm grid. The detection pattern is shown for dowels of varying diameters that are placed in front of the sensor

A 6.1-mm (0.25-inch) diameter dowel
B 2.54-cm (1-inch) diameter dowel
C 8.89-cm (3.5-inch) diameter dowel

D 11-inch wide board moved left to right with the board parallel to the front sensor face. This shows the sensor's range capability.

Note: For people detection the pattern typically falls between charts A and B.



Beam Characteristics are Approximate

Beam Pattern drawn to a 1:95 scale for easy comparison to our other products.

MB1010 Features and Benefits

- Most popular ultrasonic sensor
- Low power consumption
- Easy to use interface
- Can detect people to 8 feet
- Great balance between sensitivity and object rejection
- Can be powered by many different types of power sources

MB1010 Applications and Uses

- Great for people detection
- Security
- Motion detection
- Used with battery power
- Autonomous navigation
- Educational and hobby robotics
- Collision avoidance

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MB1020 LV-MaxSonar-EZ2

The LV-MaxSonar-EZ2 is a good compromise between sensitivity and side object rejection. The LV-MaxSonar-EZ2 is an excellent choice for applications that require slightly less side object detection and sensitivity than the MB1010 LV-MaxSonar-EZ1.

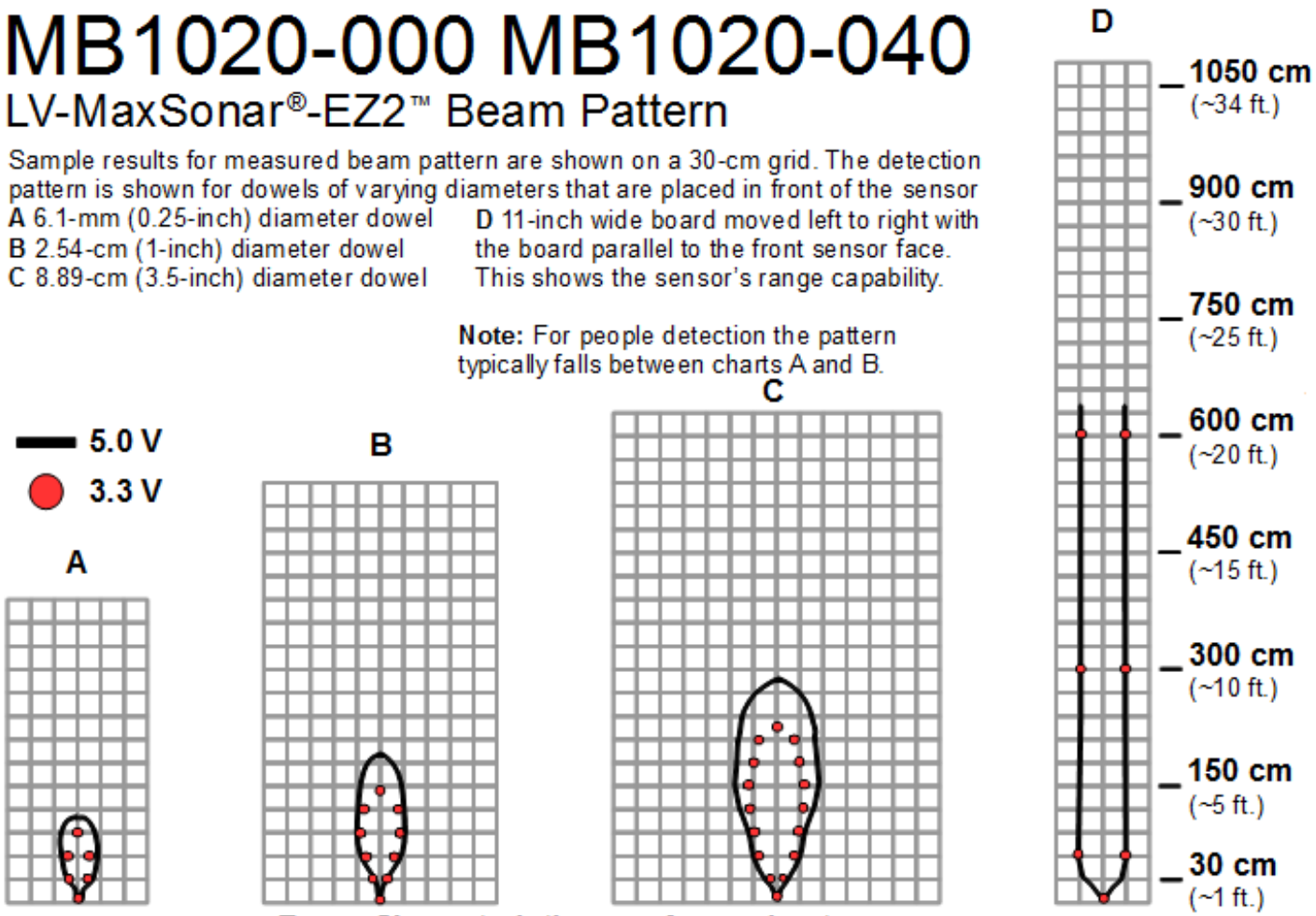
MB1020-000 MB1020-040

LV-MaxSonar®-EZ2™ Beam Pattern

Sample results for measured beam pattern are shown on a 30-cm grid. The detection pattern is shown for dowels of varying diameters that are placed in front of the sensor

A 6.1-mm (0.25-inch) diameter dowel
B 2.54-cm (1-inch) diameter dowel
C 8.89-cm (3.5-inch) diameter dowel
D 11-inch wide board moved left to right with the board parallel to the front sensor face. This shows the sensor's range capability.

Note: For people detection the pattern typically falls between charts A and B.



Beam Characteristics are Approximate

Beam Pattern drawn to a 1:95 scale for easy comparison to our other products.

MB1020 Features and Benefits

- Great for applications where the MB1010 is too sensitive.
- Excellent side object rejection
- Can be powered by many different types of power sources
- Can detect people up to approximately 6 feet

MB1020 Applications and Uses

- Landing flying objects
- Used with battery power
- Autonomous navigation
- Educational and hobby robotics
- Large object detection

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MB1030 LV-MaxSonar-EZ3

The LV-MaxSonar-EZ3 is a narrow beam sensor with good side object rejection. The LV-MaxSonar-EZ3 has slightly wider beam width than theMB1040 LV-MaxSonar-EZ4 which makes it a good choice for when the LV-MaxSonar-EZ4 does not have enough sensitivity for the application.

MB1030-000 MB1030-040

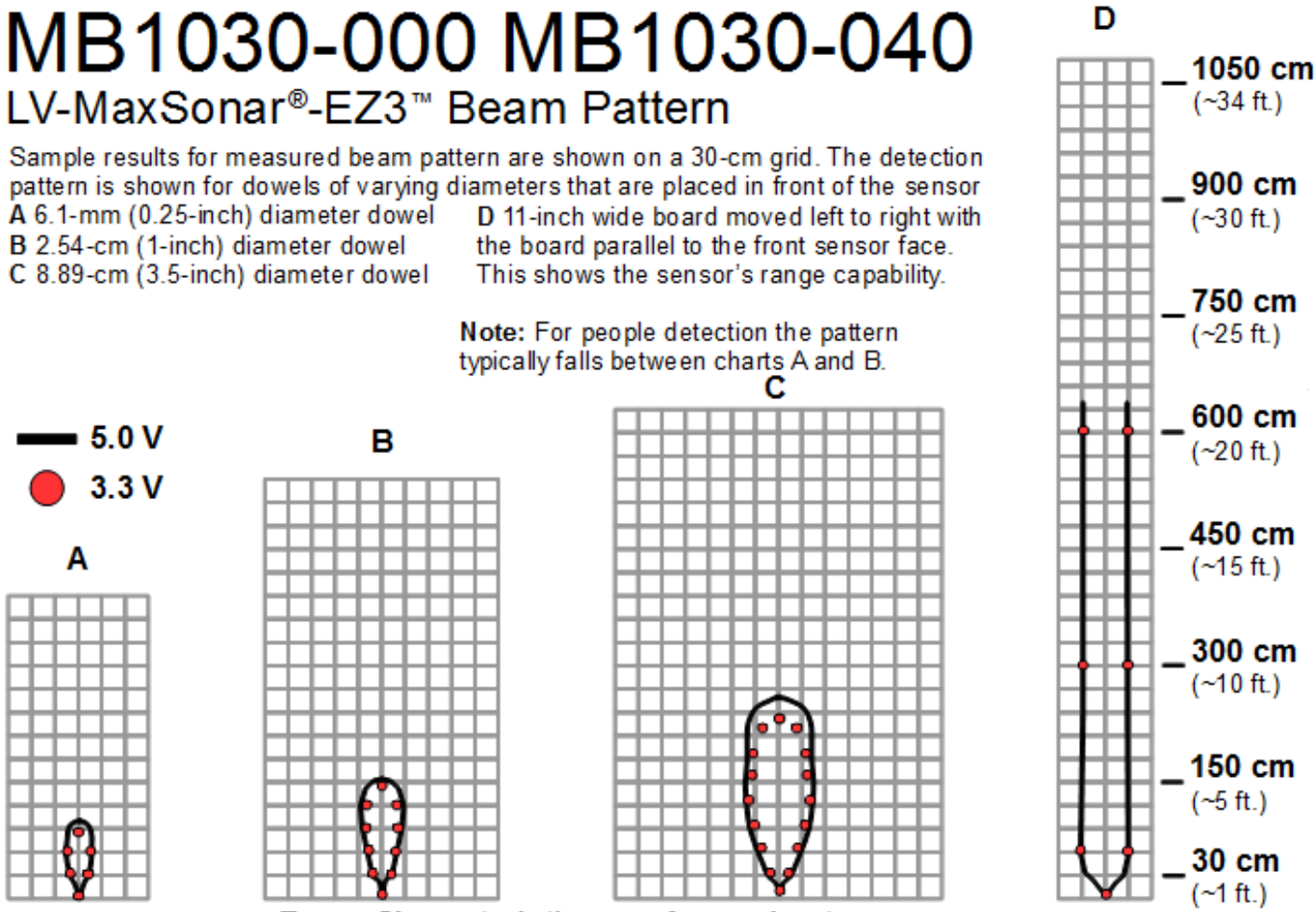
LV-MaxSonar®-EZ3™ Beam Pattern

Sample results for measured beam pattern are shown on a 30-cm grid. The detection pattern is shown for dowels of varying diameters that are placed in front of the sensor

A 6.1-mm (0.25-inch) diameter dowel
B 2.54-cm (1-inch) diameter dowel
C 8.89-cm (3.5-inch) diameter dowel

D 11-inch wide board moved left to right with the board parallel to the front sensor face. This shows the sensor's range capability.

Note: For people detection the pattern typically falls between charts A and B.



Beam Characteristics are Approximate

Beam Pattern drawn to a 1:95 scale for easy comparison to our other products.

MB1030 Features and Benefits

- Excellent side object rejection
- Low power consumption
- Easy to use interface
- Great for when MB1040 is not sensitive enough
- Large object detection
- Can be powered by many different types of power sources
- Can detect people up to approximately 5 feet

MB1030 Applications and Uses

- Landing flying objects
- Used with battery power
- Autonomous navigation
- Educational and hobby robotics

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MB1040 LV-MaxSonar-EZ4

The LV-MaxSonar-EZ4 is the narrowest beam width sensor that is also the least sensitive to side objects offered in the LV-MaxSonar-EZ sensor line. The LV-MaxSonar-EZ4 is an excellent choice when only larger objects need to be detected.

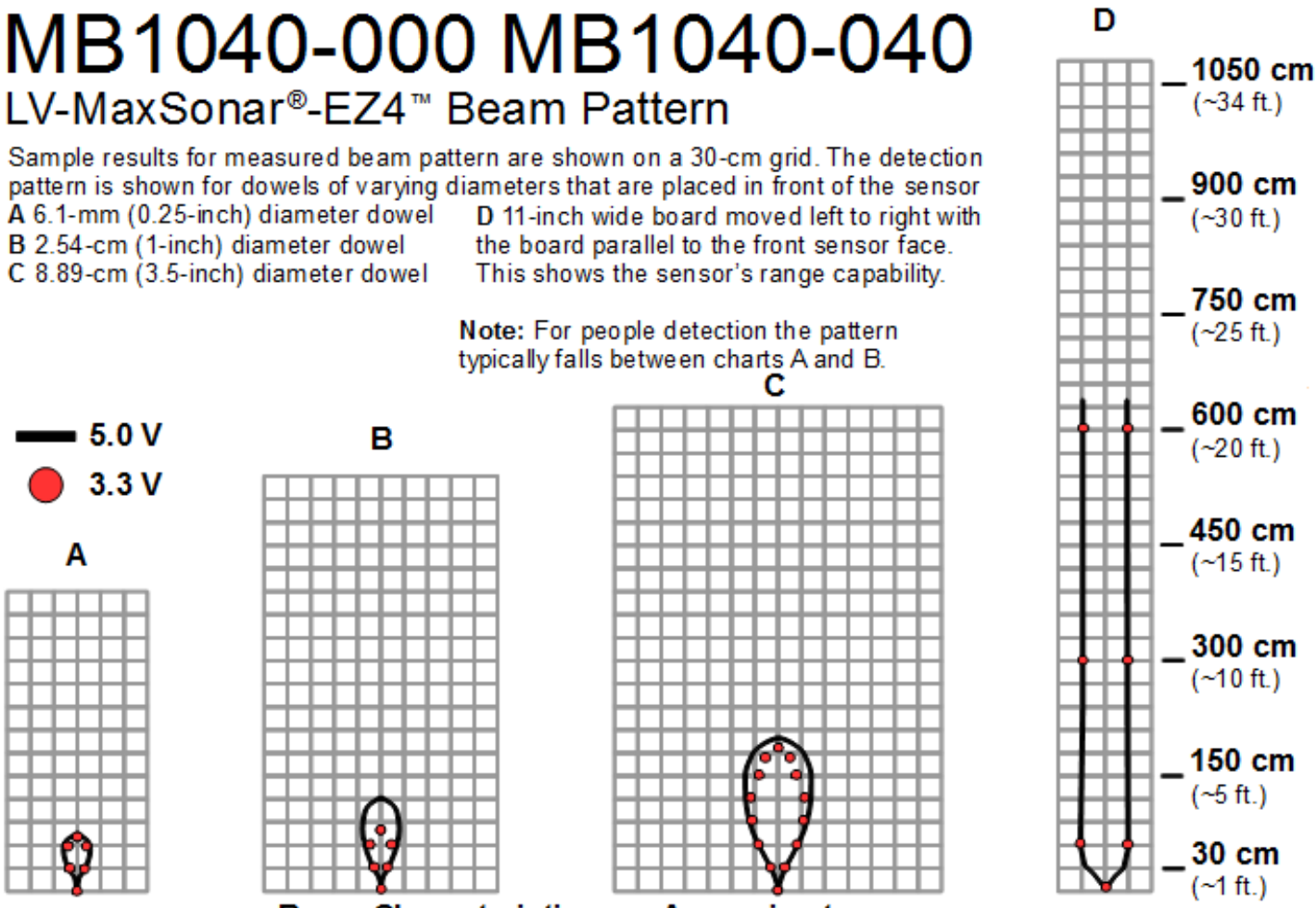
MB1040-000 MB1040-040

LV-MaxSonar®-EZ4™ Beam Pattern

Sample results for measured beam pattern are shown on a 30-cm grid. The detection pattern is shown for dowels of varying diameters that are placed in front of the sensor

A 6.1-mm (0.25-inch) diameter dowel
B 2.54-cm (1-inch) diameter dowel
C 8.89-cm (3.5-inch) diameter dowel
D 11-inch wide board moved left to right with the board parallel to the front sensor face. This shows the sensor's range capability.

Note: For people detection the pattern typically falls between charts A and B.



Beam Characteristics are Approximate

Beam Pattern drawn to a 1:95 scale for easy comparison to our other products.

MB1040 Features and Benefits

- Best side object rejection in the LV-MaxSonar-EZ sensor line
- Low power consumption
- Easy to use interface
- Best for large object detection
- Can be powered by many different types of power sources
- Can detect people up to approximately 4 feet

MB1040 Applications and Uses

- Landing flying objects
- Used with battery power
- Autonomous navigation
- Educational and hobby robotics
- Collision avoidance

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Part Numbers

All part numbers are a combination of a six-character base followed by a dash and a three-digit product code. Please review the following table for more information on the three-digit product code.



0	Not Applicable
1	3/4" NPS WR (Standard)
2	3/4" NPS WRC (Compact)
3	Ultra Compact
4	Ultra Compact Flush Mount
5	1" NPS
6	1" BSPP
7	30MM 1.5
8	Extended Horn
A	MB7974 Snap On Horn
B	90 Degree Horn
C	Infrared/Ultrasonic High Sensitivity Horn

0	No Options (Bagged)
1	F-Option
2	P-Option
3	F-Option and P-Option
4	No Options (Trayed)
5	TTL (Bagged)
6	TTL (Trayed)

0	No Options
1	7 Strand Wire Attached (MB7984) – 3ft Length
2	90 Degree 7 Pin Header
3	Straight 7 Pin Header
4	7 Pin Molex Connector
5	8 Pin M12 Connector with Cable – 3ft Length
6	7 Pin Screw Block Terminal
7	Radio Bridge Connector with Cable (MB7985) – 5ft Length
8	Custom Connector without Cable
9	Custom Connector with Cable
A	7 Strand Wire Attached (MB7984) – Custom Length
B	USB Connector with Cable (Style A) – 5M Length
C	USB Connector with Cable (Style C) – 5M Length
D	USB Connector with Cable (Style A) – 10M Length
E	USB Connector with Cable (Style C) – 10M Length
F	8 Pin M12 Connector with Cable - Custom Length
G	90 Degree Pin Connector (7 Pin)

The following table displays all of the active and valid part numbers for this product.

Active Part Numbers for MB1000, MB1010, MB1020, MB1030, MB1040, MB1060 and MB1061					
MB1000-000	MB1010-000	MB1020-000	MB1030-000	MB1040-000	MB1060-000
MB1000-040	MB1010-040	MB1020-040	MB1030-040	MB1040-040	MB1061-000

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After reviewing this datasheet, do you have any more questions?

We offer Technical Support on all of our products even if you purchased them through one of our many vendors worldwide.

You can **fill out a Technical Support form** for assistance on a sensor here --> **Technical Support**

Not sure which sensor you need for your application?

We offer Sensor Selection Assistance, click the link here to fill out a form for support --> **Sensor Selection Help**

Looking for tutorials to help you get started?

Frequently Asked Questions about Our Sensors

We receive many questions about our products and services. This resource offers answers to common inquiries we receive about our product lines and their application.

Fully Calibrated Beam Patterns

All of our sensors are factory calibrated to provide consistent beam patterns, detection zones, to fit into a wide variety of applications. In our product lines, each model number comes with a different beam pattern that reflects the sensitivity and the detection zone of how it sees a target. Additionally, we strive to maintain consistency between our finished products, and you will see little to no deviation between sensors of the same model. This allows you to have confidence in your final application when using multiple sensors.

Understanding Range Readings

The success of an application may hinge upon knowing the exact location of a target. However, a sensor may report one meter even if the target is not exactly one meter away from the sensor. Sensor specifications, such as resolution, precision, and accuracy, help you to understand sensor performance.

How to Use Multiple Ultrasonic Sensors

This guide covers three ways to run your sensors in a Multiple Sensor environment and issues you may face.

Contact us now with any questions at sensors@maxbotix.com or call +1-218-454-0766.

Please call during our preferred business hours of 8:00 am – 4:30 pm CST on Monday through Thursday and 8:00 am – 2:00 pm CST on Friday, or you may leave us a voicemail anytime.

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SEPTEMBER 1 2026

BEFORE I ADD RELAY, I SHOULD CHECK TO SEE IF I CAN HARNESS POWER FROM I²C CONNECTION TO POWER HELTEC

↳ IS THERE A POWER PIN THAT IS ALSO DISABLED?

CAN I USE STEMMA QT CONNECTOR TO POWER BUS, TURN I²C POWER OFF WITH SOFTWARE

9/2/26

HAD PROBLEMS DOWNLOADING PROGRAMS THROUGH ARDUINO SOFTWARE
- FINALLY GOT IT WORKING - COULD IT BE CABLES? POWER ONLY, NO COMMUNICATION?

TO DO: WRITE PROGRAM TO TEST POWER OFF & ON

USED STEMMA QT CONNECTORS TO CONNECT NEW BME 280 SENSOR AND ANOTHER TO CONNECT TO BUS ON PROTOTYPE BOARD

POWER OFF NEOPIXEL/I²C
GO TO LIGHT SLEEP

CHECK VOLTAGES ON BUS & OTHER FEATHER PINS
- CAN I TURN OFF MAX30101 AND HELTEC & BME 280 WITHOUT RELAY?

TEST PROGRAM

TURN OFF NEO PIXEL WITH SERIAL PROMPTS
WAIT 10 sec
LIGHT SLEEP MODE
30 sec
WAKE
10 sec
TURN ON NEOPIXEL
10 sec

9/6/26 - 9/7

SOMETHING IS WRONG. I CAN'T COMMUNICATE WITH BME280. EVEN EXAMPLE SCRIPT FROM ADAFRUIT WON'T WORK!

TRIED USING I²C PINS INSTEAD OF STEMMA QT PLUGS AND THAT DIDN'T WORK EITHER + TRYING OTHER WIRES/PLUGS

9/7

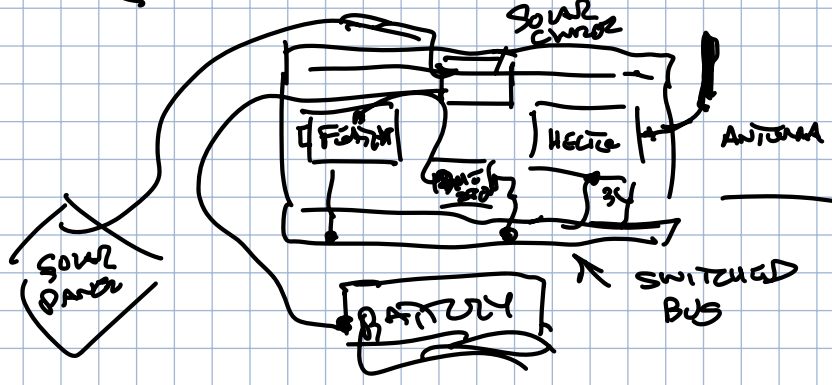
BOUGHT: 1 NEW BME280
1 NEW FEATHER ESP32 V2
1 NEW FEATHER ESP32 + BME280 COMBO!
- ADDITIONAL STEMMA QT WIRES

START OVER!

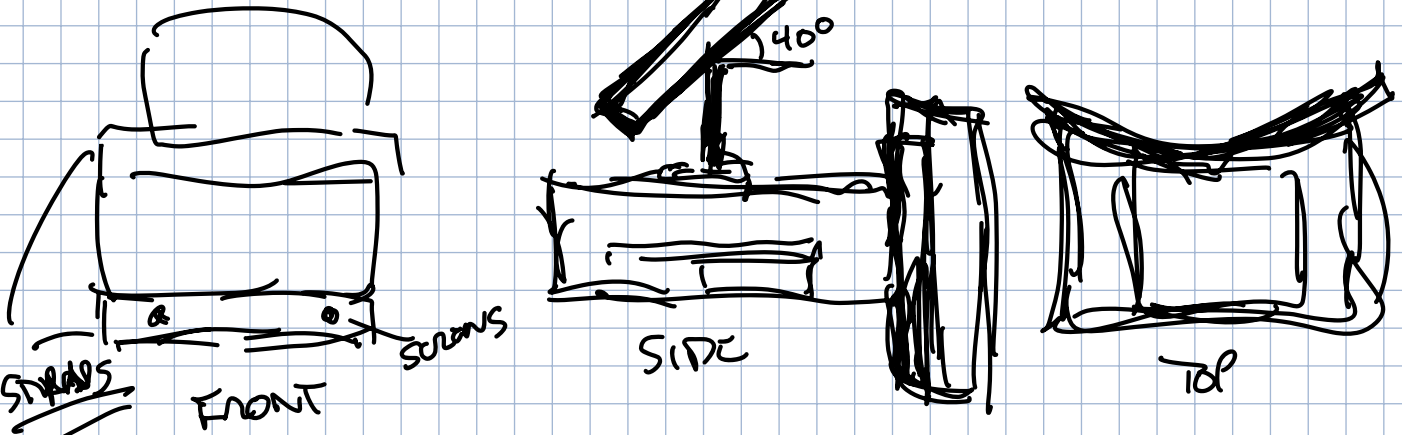
BUT I CAN TURN OFF I²C POWER BEFORE SLEEP, SO WE CAN SAVE POWER THAT WAY

- 2W SOLAR PANEL
- ADAFRUIT SOLAR CHARGER
- 2800 MWHR BATTERY (ADAFRUIT)
- FEATHER (W BME280)
- HELTEC LORA BOARD

NO NEED FOR RELAY!



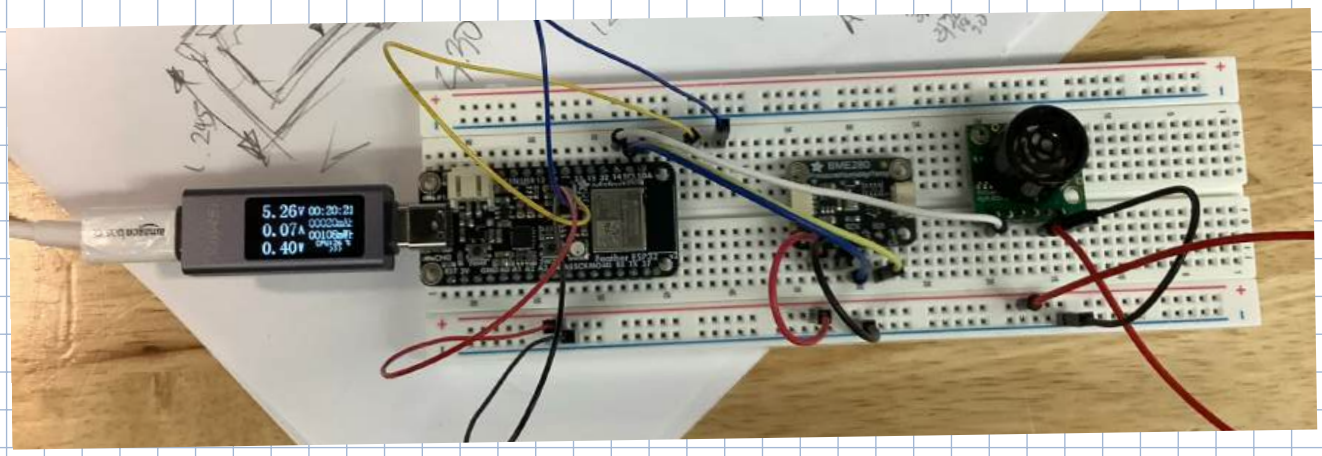
1 LONG PROTO BOARD (SOLDERABLE)



9/9/26

SLEEPTEST 2 – JUST BME 280 o FEATHER
– BME CANNOT BE FOUND!!

SLEEPTEST 3 – ADD MAXBOTIX SENSOR
(USING PIN ~~18~~ 14 TO COMMUNICATE



SLEEP 3 TEST

① BME 280 TURNS ON, BUT DOES NOT SEND DATA

② MAXBOTIX WORKS!! (ON PIN 14)

1 mAHr PER MINUTE FOR 15 sec DELAY PER CYCLE
1500 mAHr PER DAY?

POWER CALCULATION

HOW MANY mA Hr DOES THE 2W SOLAR PANEL GENERATE?

$1\ \text{mA Hr} = 36$ COULOMBS — A MEASUREMENT OF CHARGE

$1 \times 10^{-3} \frac{C}{s} \cdot 3600\ s = 3.6\ C$

$2\ W = 2\ A \cdot V$

CURRENT IS $\frac{W}{V} = \frac{2\ A \cdot V}{5\ V} = .4\ A$

$400\ mA$

CHARGE IS $\frac{1}{6} \cdot 24\ Hr$ GUESS 4 FULL HOURS A DAY

CHARGE GENERATED $= 400\ mA \cdot 4\ Hr$

$= 1600\ mA\ Hr / DAY$

BATTERY HAS 2200 mA Hr CAPACITY

IF THE SENSOR SLEEPS FOR 14 MINUTES AND RUNS FULL THROTTLE FOR 1 MINUTE, THEN CONSUMPTION IS:

$$\left[14\ \text{min} \cdot 1\ mA \cdot \frac{1\ Hr}{60\ \text{min}} + 1\ \text{min} \cdot 100\ mA \cdot \frac{1\ Hr}{60\ \text{min}}\right] \cdot 24\ Hr \cdot 4\ \frac{\text{cycles}}{Hr} / DAY$$

(1 mA: LIGHT GUESS; 100 mA: FULL UP GUESS)

$$\frac{14}{60} \cdot 24 \cdot 4\ \frac{mA\ Hr}{DAY}$$

$$\frac{114 \cdot 6 \cdot 4 \cdot 4}{6 \cdot 10}$$

$$\begin{array}{r} 184 \\ 16 \\ \hline 684 \\ 114 \\ \hline 1824 \\ 10 \end{array} = 182\ mA\ Hr / DAY$$

IS THAT ALL?

HOW ABOUT 8 MIN OFF 2 MIN ON ?

$$\frac{8 + 200 \cdot \frac{24.6}{60}}{} = 24.6$$

TOO SMALL?

$200 \cdot 1.6$

< 500 MA/DAY!!

MINUTES IN A DAY

$60 \cdot 24 =$

$$\begin{array}{r} 24 \\ \times\ 6 \\ \hline 1440 \end{array}$$

1440 MINUTES ?

USE COULOMBS/DAY INSTEAD

$$\left[\left(14 \text{ MIN} \cdot 1 \cdot 10^{-3} \frac{C}{S} \cdot \frac{60 S}{MIN}\right) + \left(1 \text{ MIN} \cdot 100 \cdot 10^{-3} \frac{C}{S} \cdot \frac{60 S}{MIN}\right)\right]$$

COULOMBS/CYCLE (96 CYCLES IN A DAY)

$$(14 + 100) \times 10^{-3} \cdot 60 = 60 \cdot 114 \cdot 10^{-3} = 6.84 \text{ C/CYCLE}$$

$$\begin{array}{r} 114 \\ \times\ 60 \\ \hline 6840 \end{array}$$

4 CYCLES/HR · 24 HR/DAY = 96 $\frac{\text{CYCLES}}{\text{DAY}}$

$$6.8 \frac{C}{\text{CYCLE}} \cdot 96 \frac{\text{CYCLES}}{\text{DAY}} = 680 \frac{C}{\text{DAY}}$$

FROM TOP:

$$\frac{1 \text{ MA HR}}{3.6 \text{ C}} \cdot 680 \frac{C}{\text{DAY}}$$

$$\begin{array}{r} 160 \\ 3.6 \overline{)\ 680.0} \\ 36 \\ \hline 320 \\ 216 \\ \hline 40 \end{array}$$

$160 \frac{\text{MA HR}}{\text{DAY}}$

FROM CURRENT MEASUREMENT OF DEVICES:

~ 1 mA Hr EVERY MINUTE

60 · 24 · 1 mA Hr/MIN
MIN/DAY

1400 $\frac{\text{mA Hr}}{\text{DAY}}$

OFF BY FACTOR OF 10 !!

OK!

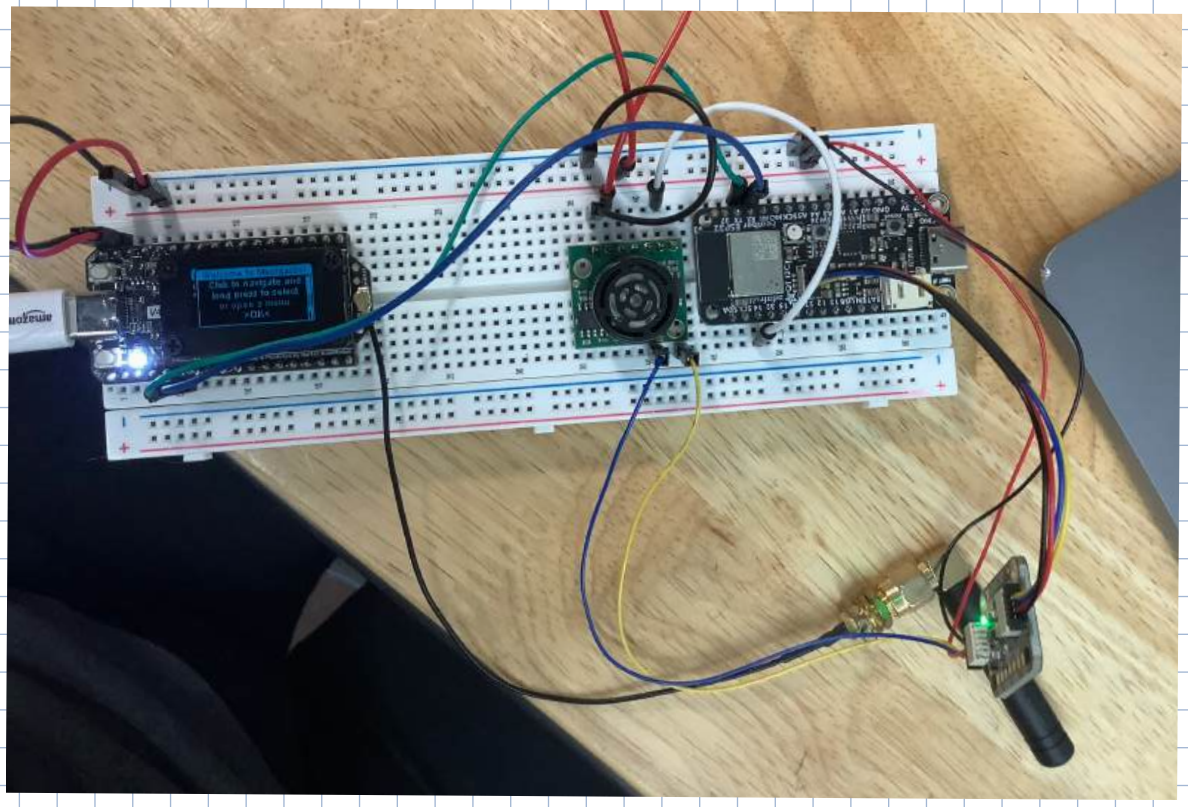
THIS MIGHT WORK!!

AH! BUT TEST SOFTWARE GOES TO SLEEP & WAKES EVERY (5-20 SECONDS!

OK! I INSTALLED NEW BME280 SENSOR AND THINGS ARE WORKING (MOSTLY)*! THE OLD BME280 MUST BE BAD!

BOUGHT FROM ADAFRUIT WITH HEADER

* SOFTWARE STILL REPORTS BME DID NOT START, BUT I DO GET T, P, H READINGS.



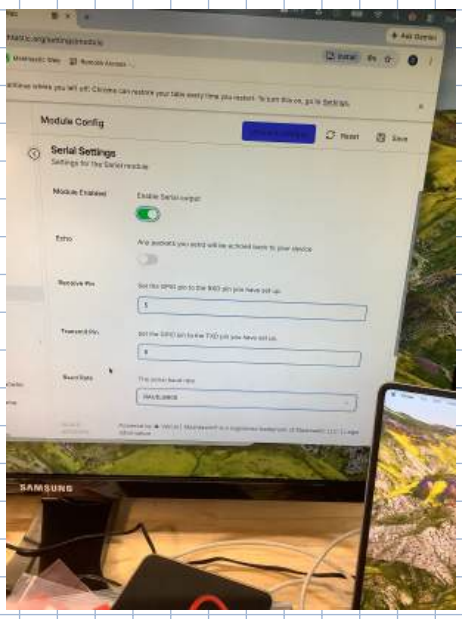
9/13 STATUS

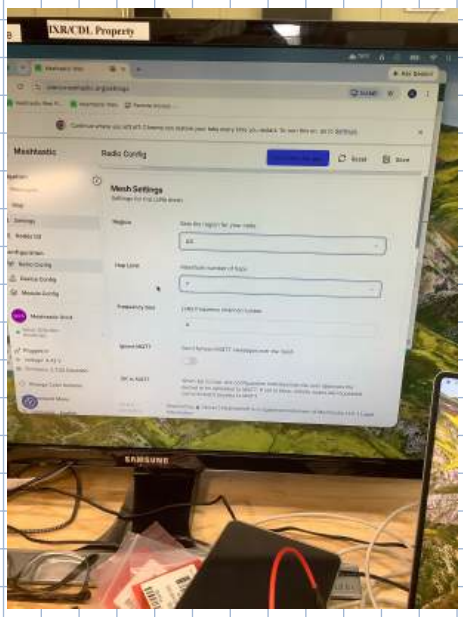
MAGNETIC SENSORS,

BME 280

FEATHER ESP32 V2,

HELTEC V4





CONFIGURING HELTEC TO ACCEPT SERIAL DATA AND THEN TEXT OUT DATA WITH LORA

9/15 UPDATE

MESHTASTIC TEXTING DOESN'T WORK

I CHECKED THE IO PORTS + SERIAL1 SOFTWARE, SO UNLESS FEATHER HARDWARE (OR HELTEC HARDWARE) IS AT FAULT, THEN COULD SOFTWARE BE PROBLEM?

IF IT IS FEATHER HARDWARE, THEN I BOUGHT A NEW ONE AND I CAN SWAP IT OUT. I AM HAVING PROBLEMS SOLDERING THOUGH + I HAD A HARD TIME SOLDERING BME280 + SOLDER WOULD NOT WET PINS.

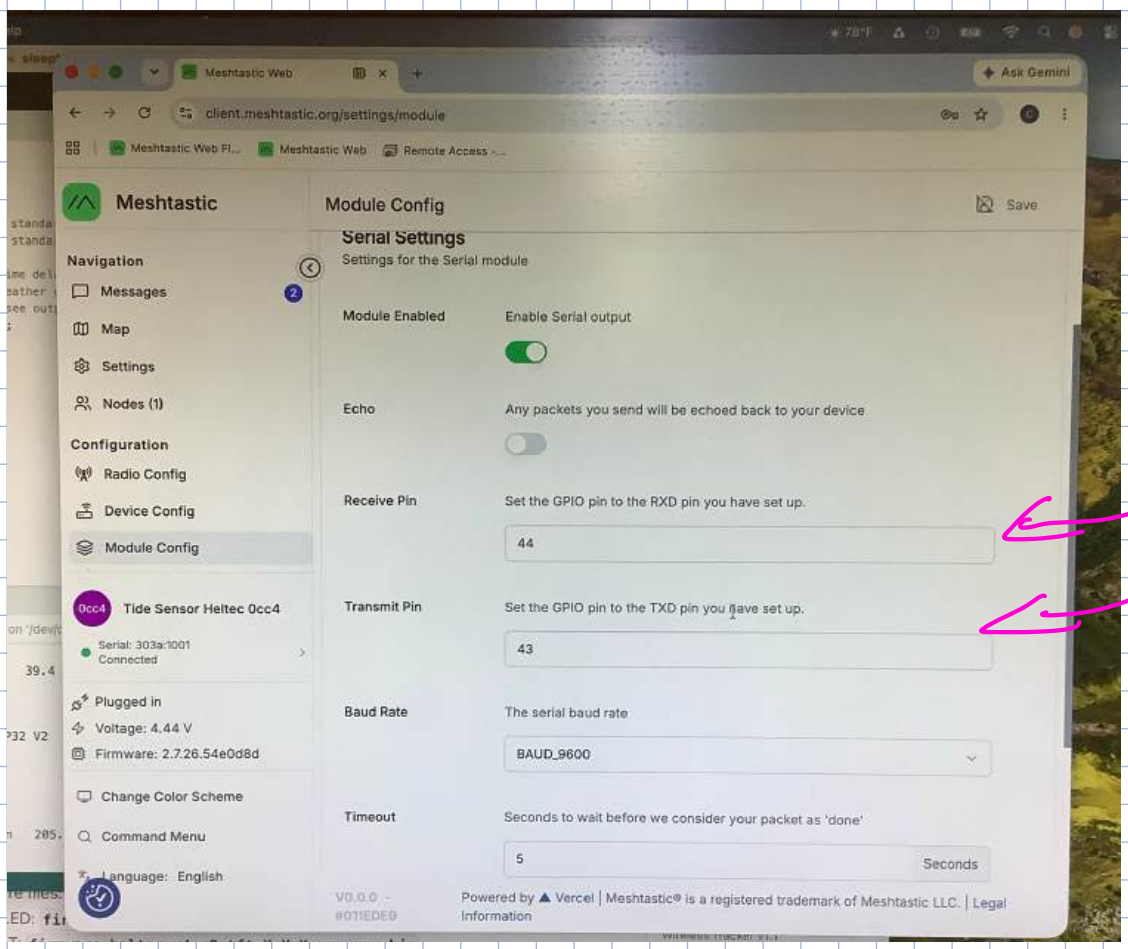
AS IT WAS LOOKING FOR A HARDWARE REPLACEMENT FOR HELTEC, I NOTICED THAT THERE ARE 2 VERSIONS, V4 AND V4-R8. THE BOARD "INK" IS APPARENTLY IDENTICAL, BUT PINOUTS ARE DIFFERENT. IT COULD BE THAT I FLASHED THE WRONG MESHTASTIC SOFTWARE ONTO THE CHIP. WEB SAYS IF WRONG VERSION IS USED, DISPLAY MAY NOT WORK AND RADIO WILL NOT WORK!

HELTEC DISPLAY WORKS WHEN ITS HOOKED UP TO USB C PORT, BUT DOES NOT WORK WHEN USING POWER FROM BUS! (BUT AN LED LIGHTS UP, SO SOME POWER IS GETTING THROUGH.)

NEXT STEP IS TO SYSTEMATICALLY INSTALL MESHTASTIC SOFTWARE - BOTH VERSIONS - AND TEST RESPONSE

IF THAT DOESN'T HELP TIME TO SOLDER PINS TO NEW FEATHER, THEN SWAP THEM TO CHECK FOR HARDWARE ISSUES.

SOFTWARE FIXES:
LIGHT SLEEP DOESN'T SEEM TO HELP. I STILL NEED TO RE-INITIALIZE EVERYTHING. MAYBE ITS BETTER TO GO INTO DEEP SLEEP AND START FROM SCRATCH



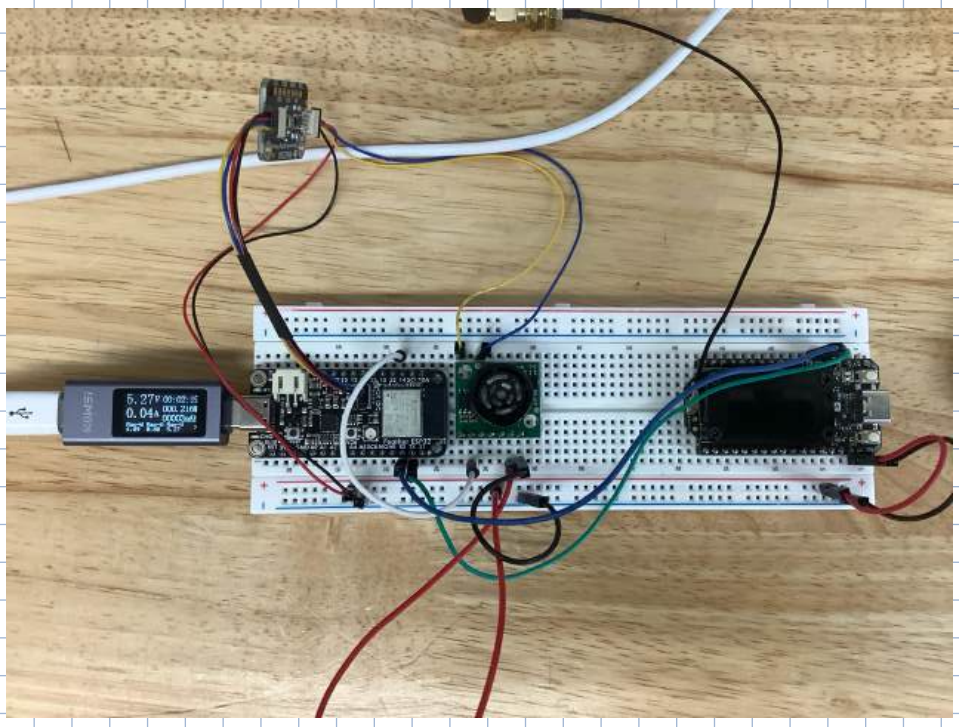
HERE WAS THE PROBLEM! PHYSICAL NOT PIN #S BUT GPIO PIN NUMBERS ARE RECOMMENDED HERE!

GPIO 44 RXD!
GPIO 43 TXD!

INCREASED WAIT TIMES AFTER TURNING ON DEVICES - SEEMS TO WORK WELL

SET SLEEP TIME TO 10 MINUTES AND PUT POWER METER IN PLACE

OK! HELTEC IS NOW BROADCASTING TIDE DATA!



2:11 PM ZOOMED POWER MEASUREMENT

3:57 PM

1 Hr 47 MIN

109 mAHr

~1500 mAHr/DAY

WITH 10 MIN INTERVALS

CLOSE TO ESTIMATED SOLAR PANEL SUPPLY ~1600 mAHr/DAY (PRETTY CONSERVATIVE ASSUMING 4 HR FULL POWER)

SLEEP TESTS

↳ INCREASE SLEEP TO 15 MIN AND I THINK WE SHOULD BE OK

NOW IT'S TIME TO THINK ABOUT PACKAGING

2Hr 125 mAHr

125
12 (12Hr PERIODS/DAY)
250
125
1500 mAHr/DAY

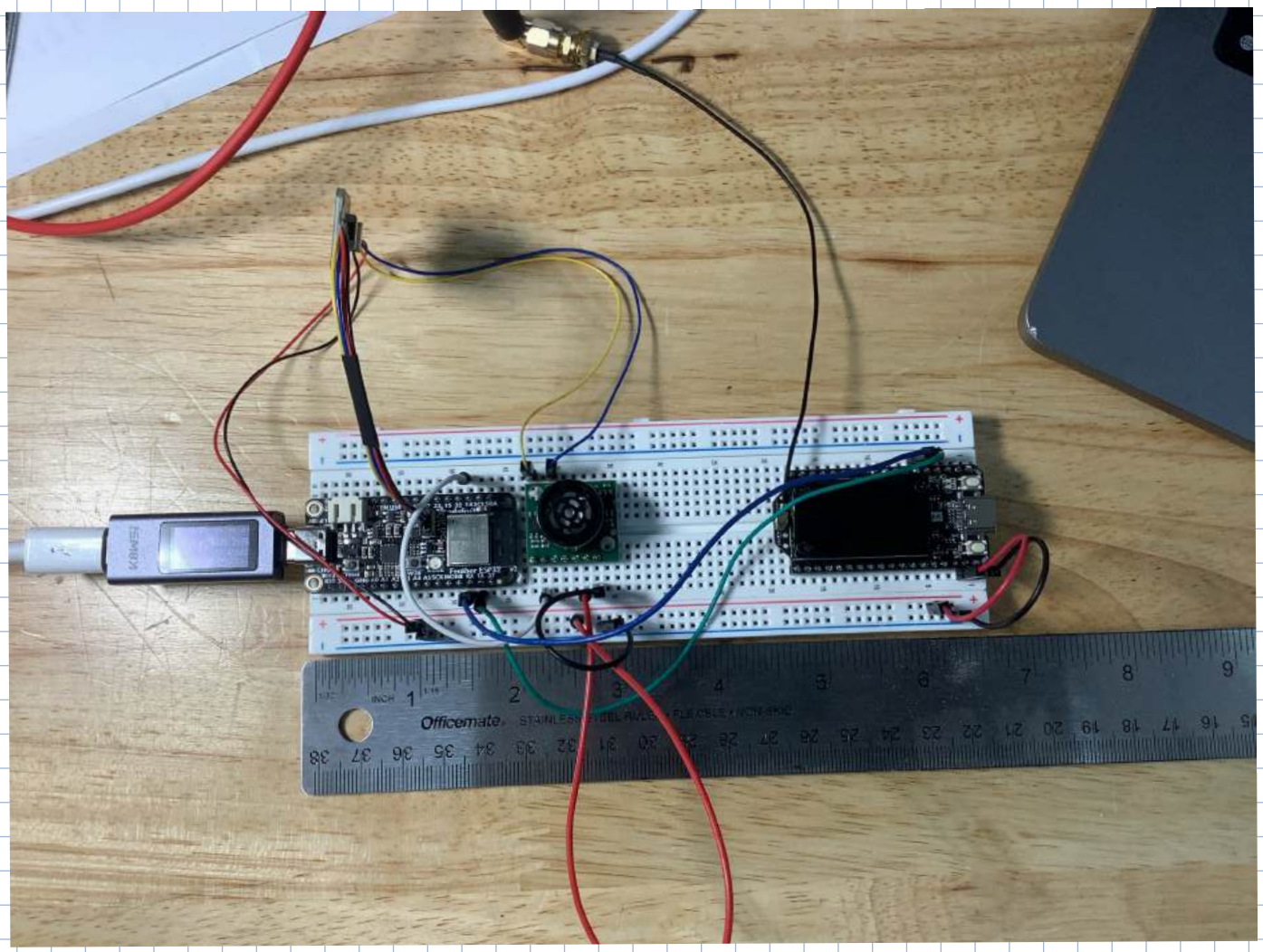
IF SET TO 15 MIN INSTEAD OF 10 MIN FOR MEASUREMENTS THEN

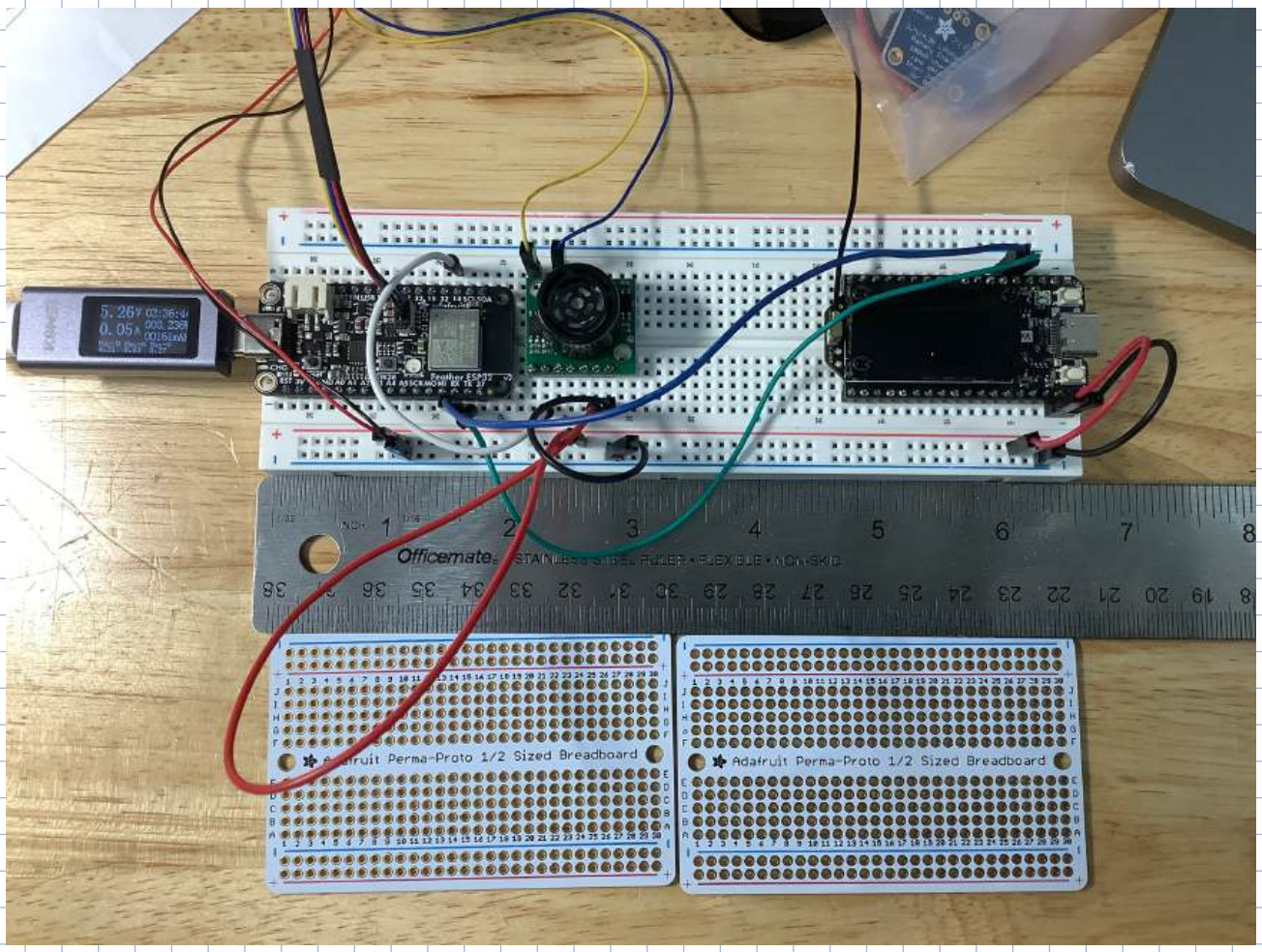
1500 · 4 Hr / 6 Hr = 1000 mAHr

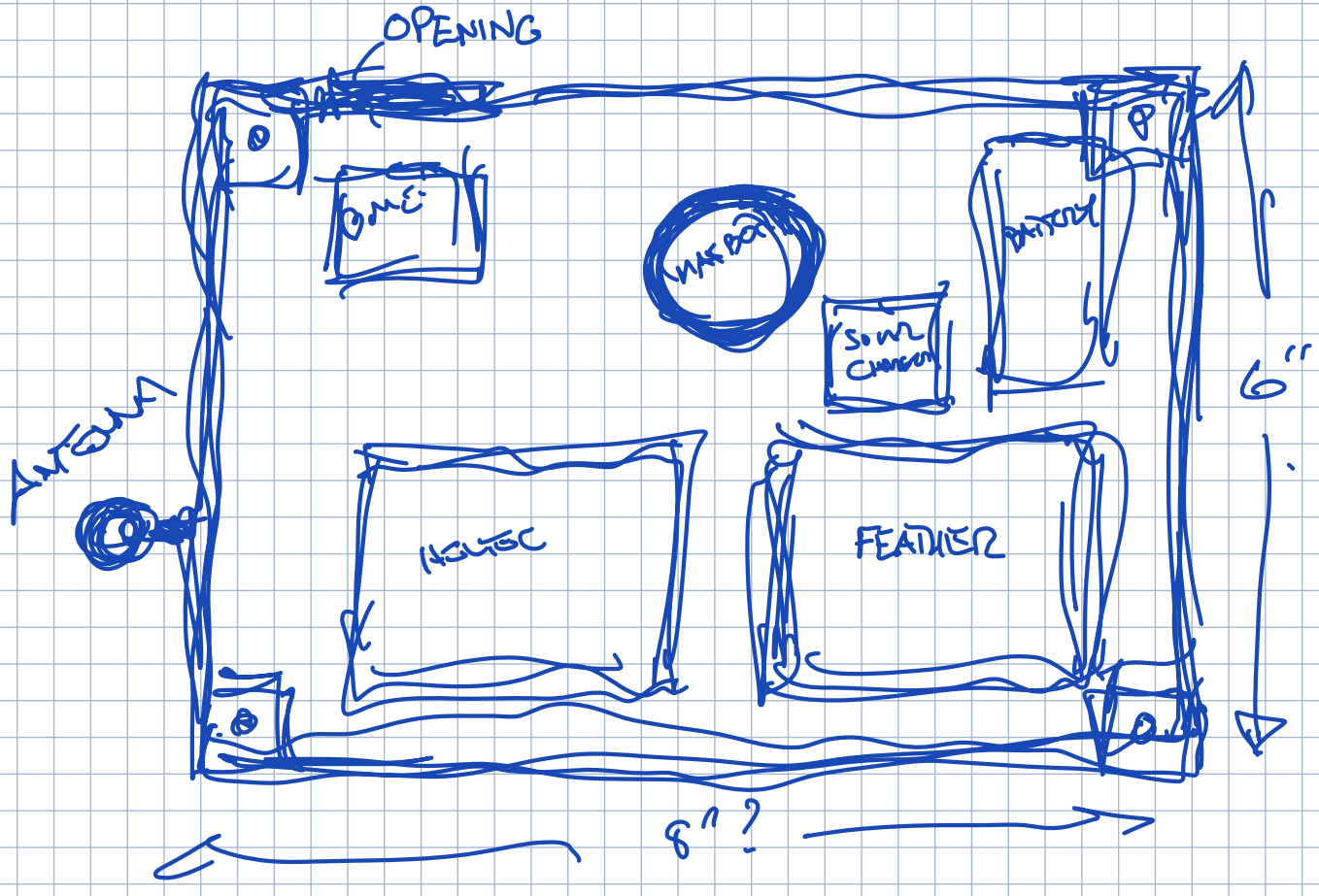
OK! 1000 mAHr/DAY SOUNDS REASONABLE

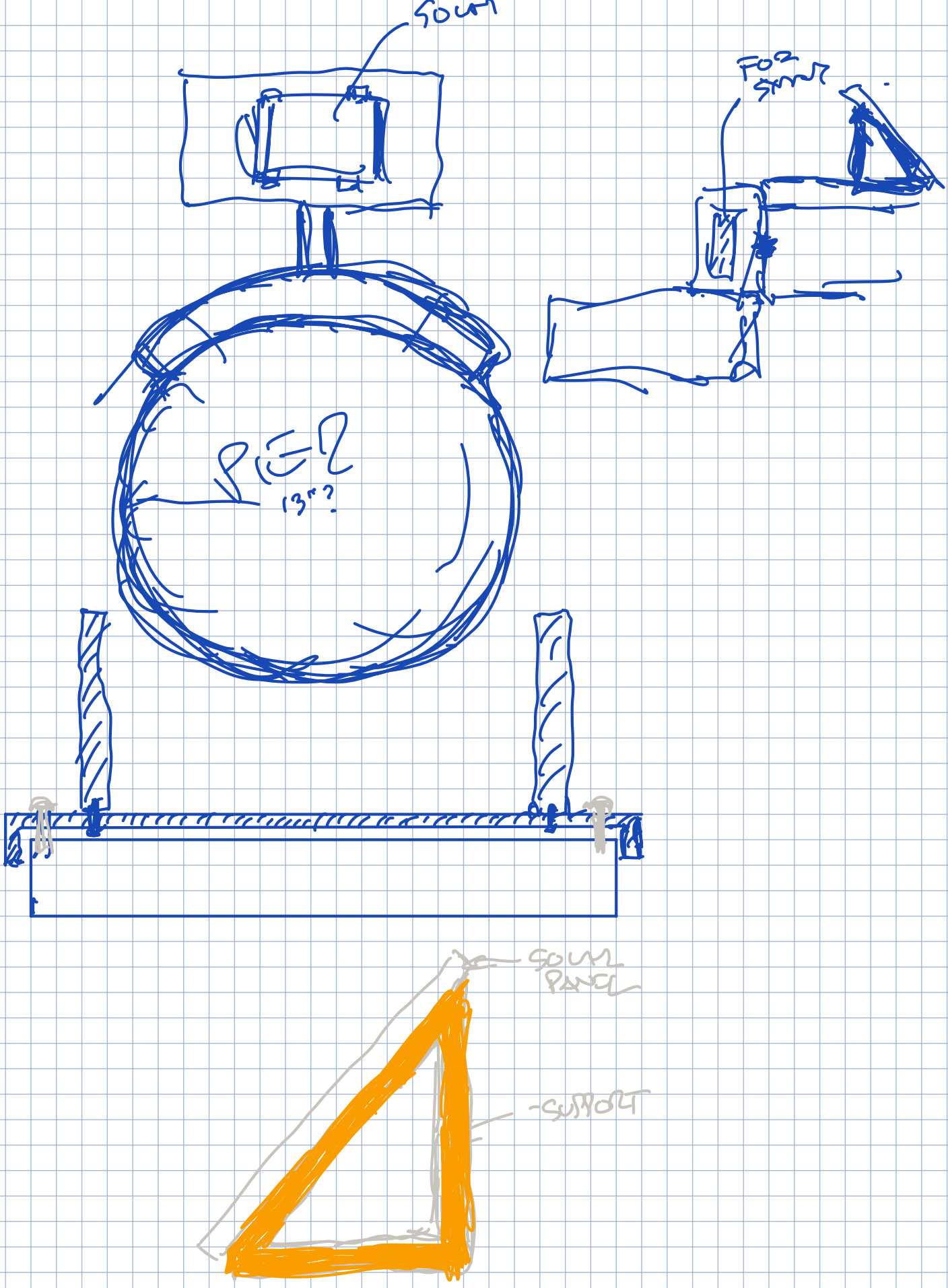
SLEEP TEST 5

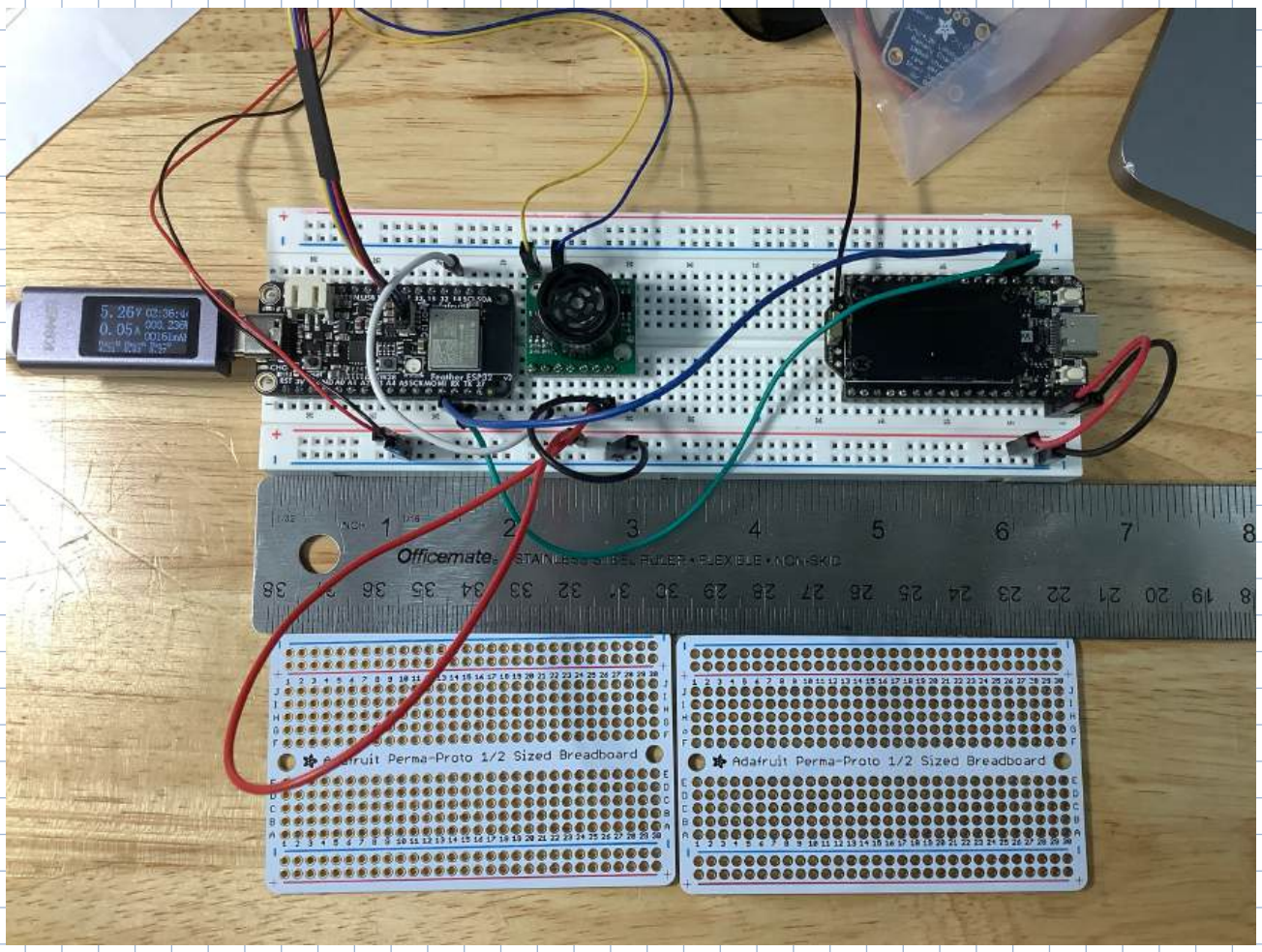
1. DEEP SLEEP (3) 0 MINUTES — OR 15 MINUTES?
2. INCREASED TIMES TO ALLOW THINGS TO WAKE UP AND TO COMPLETE PROCESSES BEFORE SLEEPING

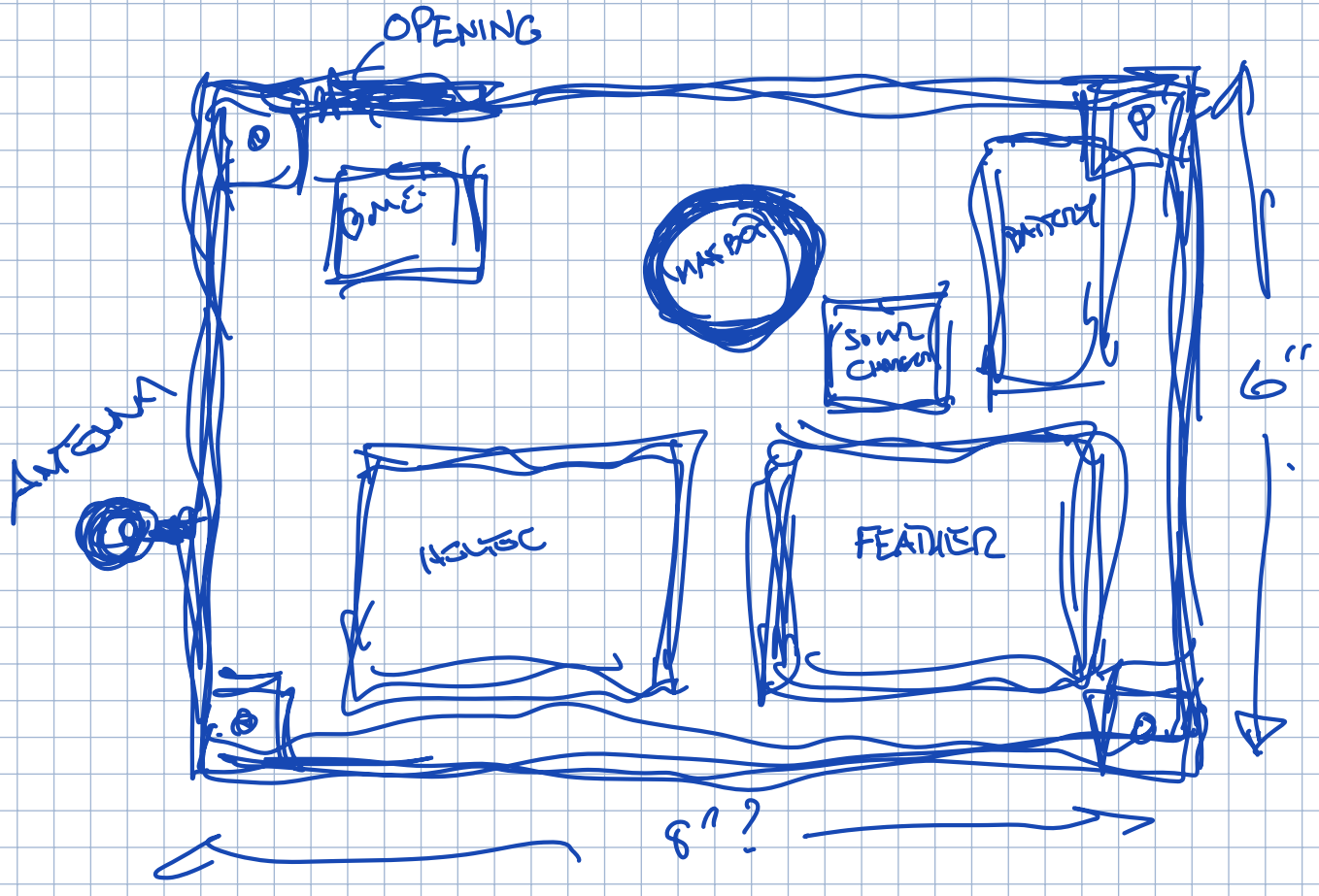


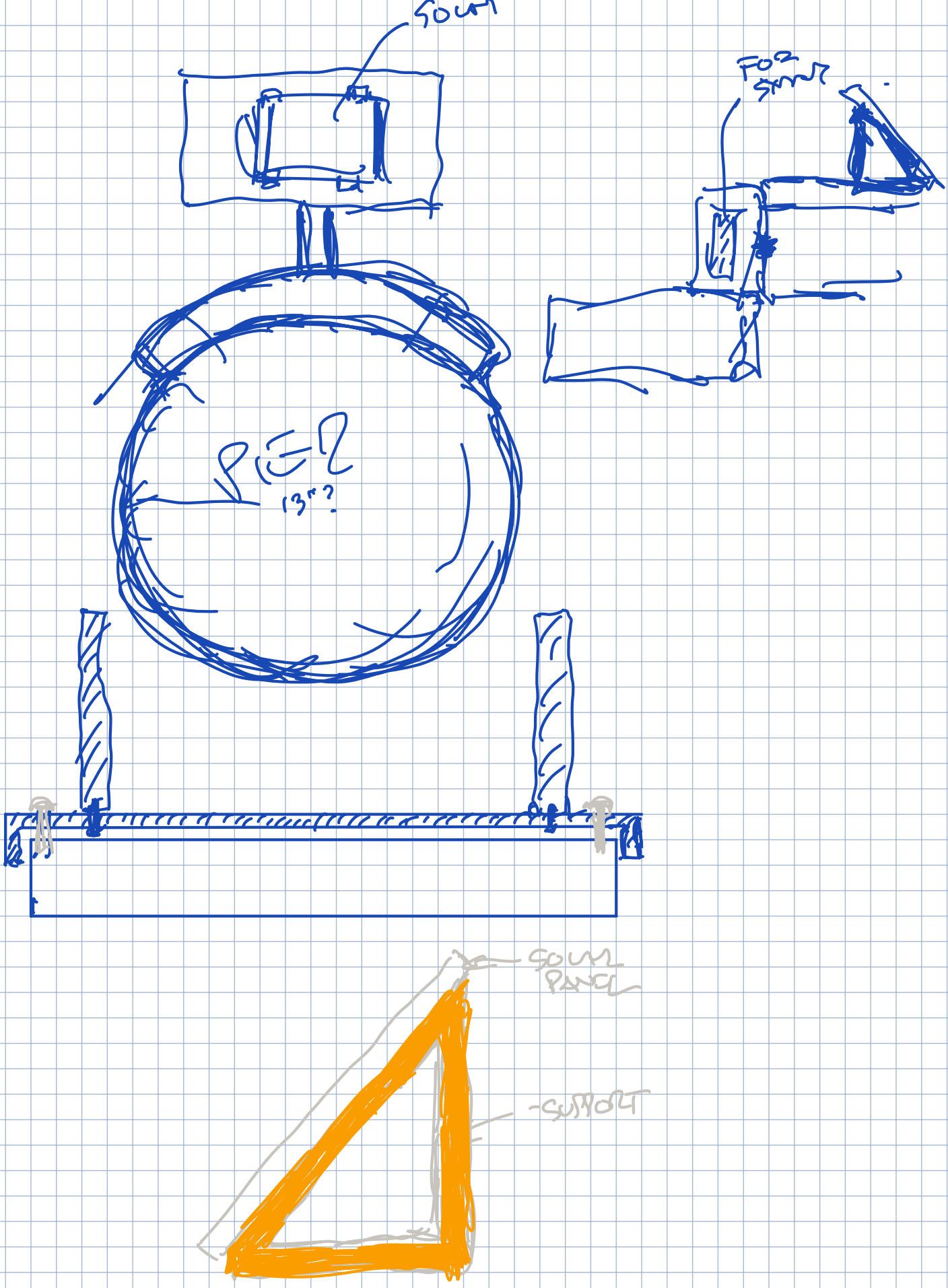

Officemate
Adafruit Perma-Proto 1/2 Sized Breadboard
Adafruit Perma-Proto 1/2 Sized Breadboard


OPENING
INK BOTTLE
6"
FEATHER


SOUND
FOR
SPACE
R=?
13"?
SOUND
PANEL
-SUPPORT


Officemate
Adafruit Perma-Proto 1/2 Sized Breadboard
Adafruit Perma-Proto 1/2 Sized Breadboard


OPENING
INK BOTTLE
6"
FEATHER


SOUND
FOR
SPACE
R=?
13"?
SOUND
PANEL
-SUPPORT